

Development and evaluation of prediction models: pitfalls and solutions

Statistics in practice I

Maarten van Smeden, PhD

Biometrisches Kolloquium

16 March 2021



UMC Utrecht
Julius Centrum

Aims of the lectures

- To give our views on the state of the art of clinical prediction modelling
- To highlight common pitfalls and potential solutions when developing or evaluating clinical prediction models
- To argue the need for less model development and more model evaluation
- To plead for increased attention to quantifying differences in model performance over time and place

Agenda

- PART I (11:00 – 12:40)
 - State of the medical prediction modeling art
 - Just another prediction model
 - Methods against overfitting
 - Methods for deciding on appropriate sample size
- PART II (13:30 – 15:10)
 - Model performance and validation
 - Heterogeneity over time and place: there is no such thing as a validated model
 - Applied example: ADNEX model
 - Future perspective: machine learning and AI

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State of the medical prediction modeling art

What do we mean with a prediction model?

“... summarize the effects of predictors to provide **individualized predictions** of the absolute risk of a **diagnostic or prognostic outcome**.”

Steyerberg, 2019

“Model development studies aim to derive a prediction model by **selecting the relevant predictors** and **combining them** statistically into a multivariable model.”

TRIPOD statement, 2015

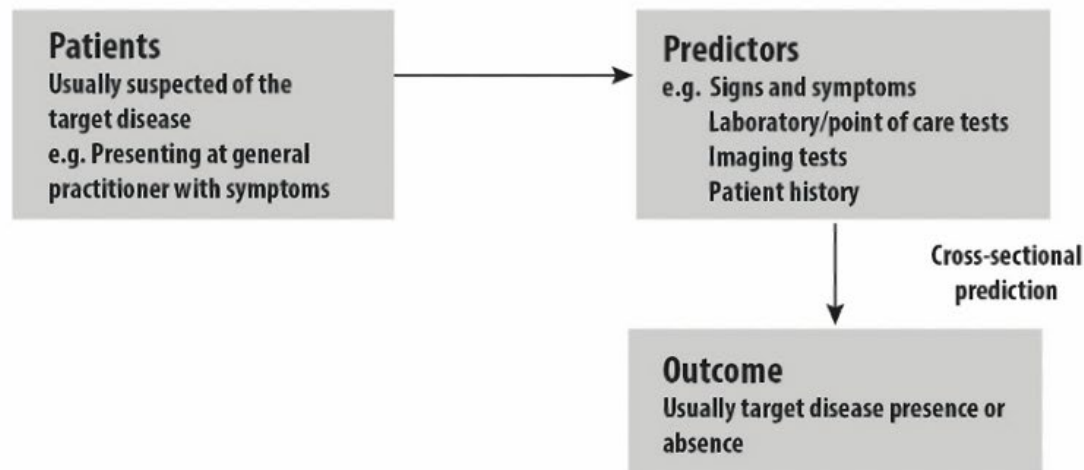
“... may be developed for **scientific or clinical reasons, or both**”

Altman & Royston, 2000

What do I mean with a prediction model?

- Mathematical rule or equation derived from empirical data describing the relation between a health outcome (Y) and input (X)
- The aim is to find a function $\hat{f}(X)$ that yields accurate individual level **predictions**, $\hat{y}_i$
- Finding the optimal $\hat{f}(X)$ is usually **not only prediction error minimization** problem, but also takes into account requirements regarding transparency, transportability, costs, sparsity etc.
- Interpretation and contribution of **individual components** in $\hat{f}(X)$ are not of *primary* interest (exception: issues related to fairness)

Diagnostic prediction modeling



Prognostic prediction modeling

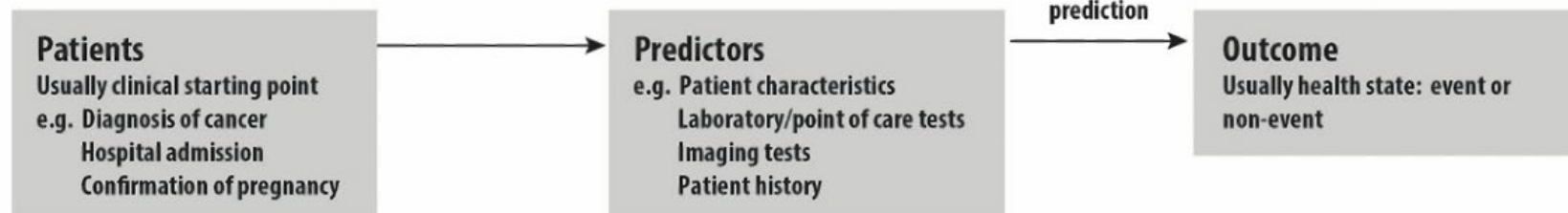


Figure: Schematic representation of diagnostic and prognostic model studies, adapted from TRIPOD reporting guideline¹

APGAR score: the oldest prediction model?

*Method of Scoring in Evaluation of Newborn Infant**

Sign	Score†		
	0	1	2
Heart rate	Absent	Slow (<100)	>100
Respiratory effort ..	Absent	Weak cry; hypoventila- tion	Good; strong cry
Muscle tone	Limp	Some flexion of extremities	Well flexed
Reflex irritability (response of skin stimulation to feet)	No response	Some motion	Cry
Color	Blue; pale	Body pink; extremities blue	Completely pink

* Evaluation 60 seconds after complete birth of infant (disregarding the cord and placenta).

† Score of 10 indicates infant in best possible condition.

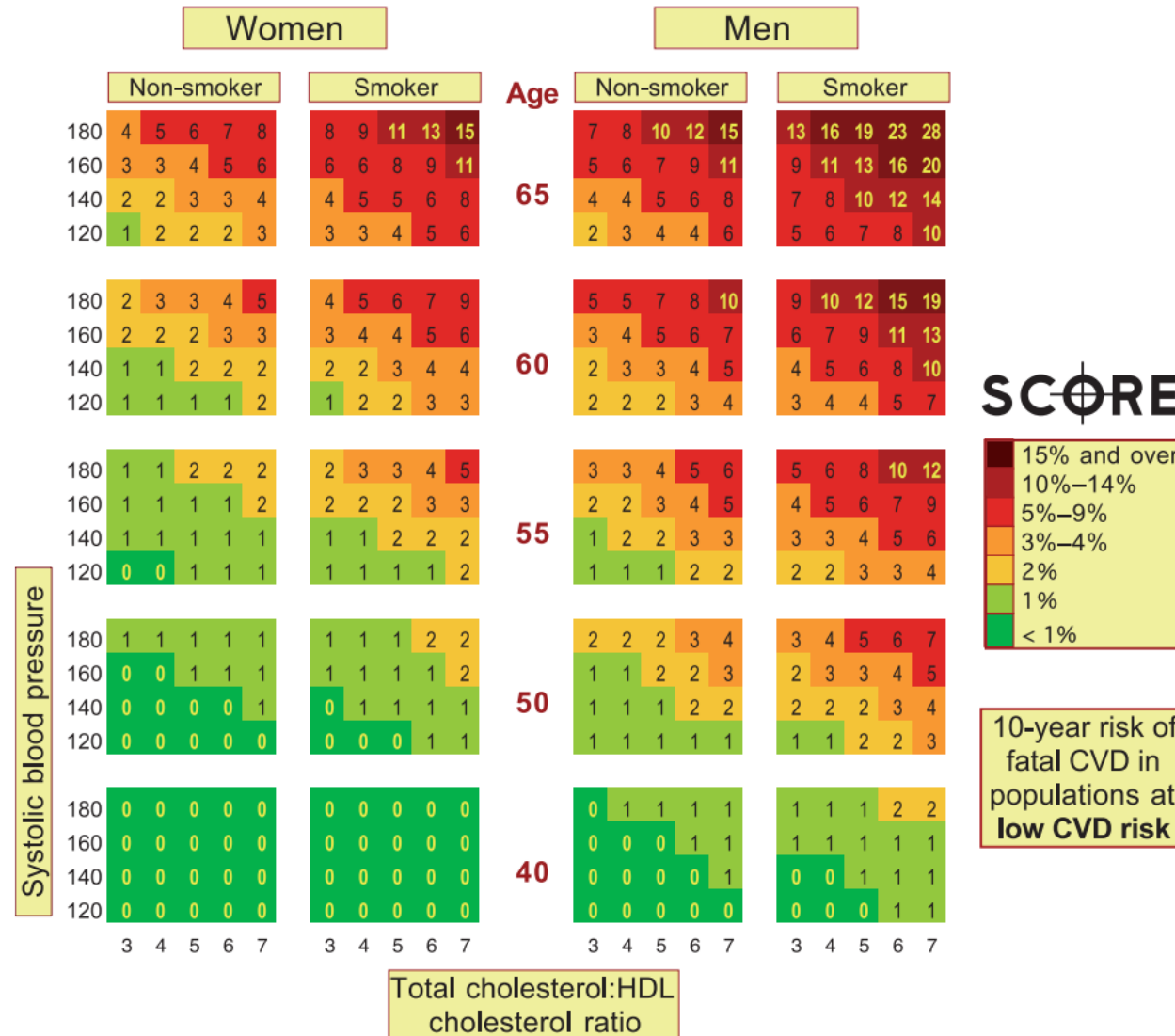
Diagnostic prediction model

Wells Prediction Rule for DVT

Clinical variable	Score
Active cancer (treatment ongoing or within previous 6 months, or palliative)	1
Paralysis, paresis, or immobilization of lower extremity	1
Bedridden for > 3 days or major surgery within previous 12 weeks requiring general or regional anesthesia	1
Localized tenderness along distribution of deep veins	1
Swelling of entire leg	1
Calf swelling at least 3 cm larger than asymptomatic leg (10 cm below tibial tuberosity)	1
Pitting edema confined to symptomatic leg	1
Collateral superficial veins (non-varicose)	1
Previously documented DVT	1
Alternative diagnosis as least as likely as DVT	-2
Clinical risk (probability)	Total
High	>2
Moderate	1-2
Low	<1

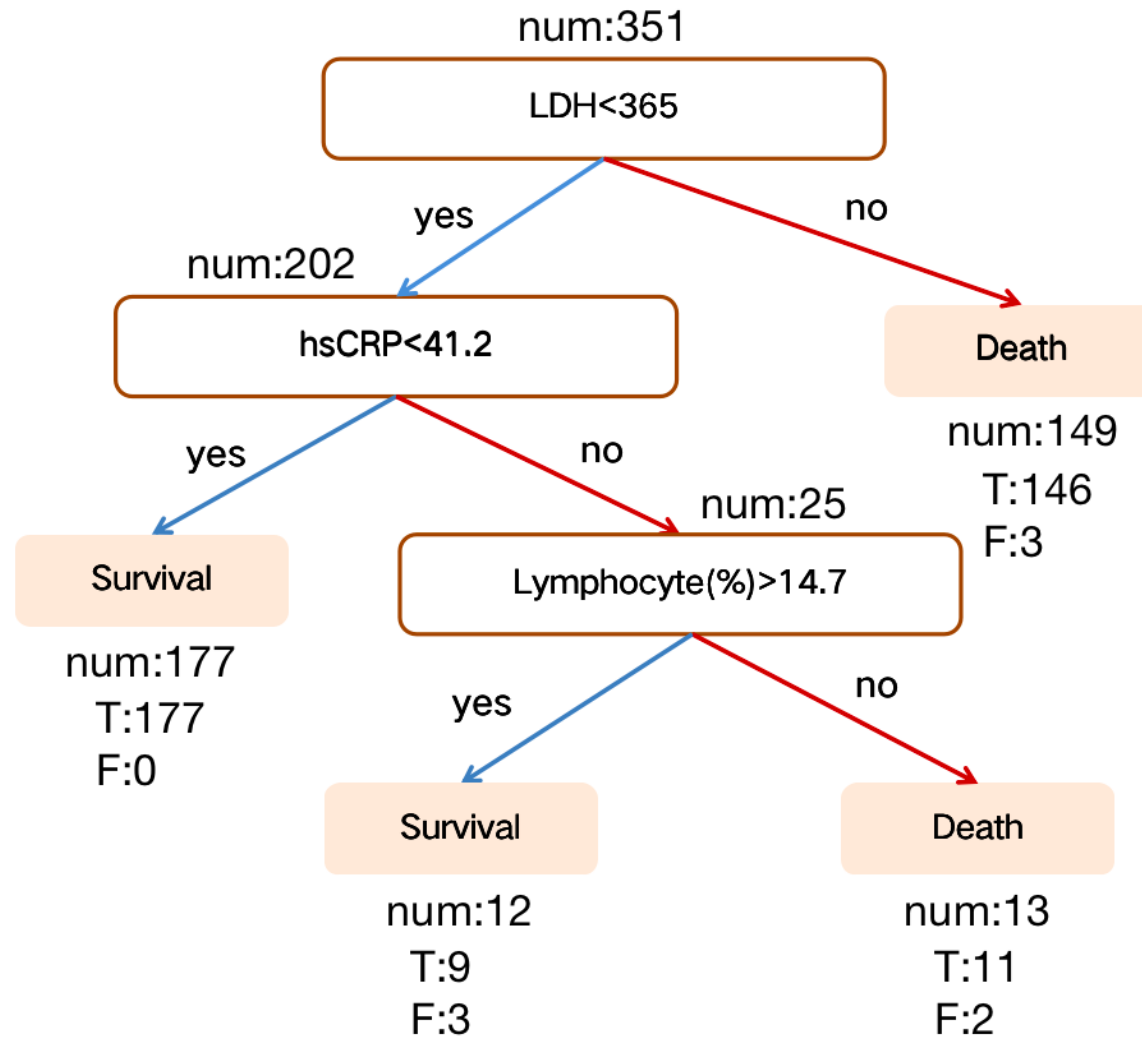
Wells et al., Lancet, 1997

Prognostic prediction model



Conroy et al. EHJ, 2003.

Simple model



Complex model

Research

JAMA | **Original Investigation** | INNOVATIONS IN HEALTH CARE DELIVERY

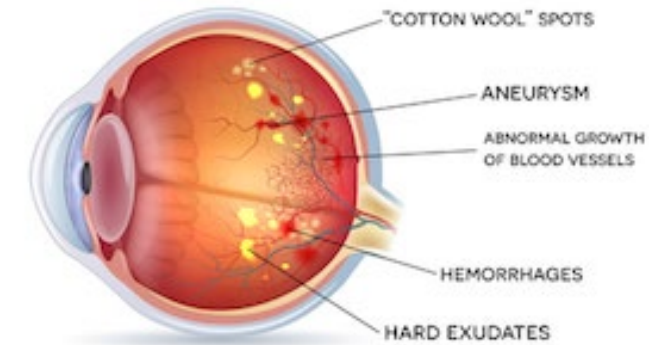
Development and Validation of a Deep Learning Algorithm for Detection of Diabetic Retinopathy in Retinal Fundus Photographs

Varun Gulshan, PhD; Lily Peng, MD, PhD; Marc Coram, PhD; Martin C. Stumpe, PhD; Derek Wu, BS; Arunachalam Narayanaswamy, PhD; Subhashini Venugopalan, MS; Kasumi Widner, MS; Tom Madams, MEng; Jorge Cuadros, OD, PhD; Ramasamy Kim, OD, DNB; Rajiv Raman, MS, DNB; Philip C. Nelson, BS; Jessica L. Mega, MD, MPH; Dale R. Webster, PhD

Deep learning (= Neural network)

- 128,000 images
- Transfer learning (preinitialization)
- Sensitivity and specificity > .90
 - Estimated from training data

Diabetic retinopathy

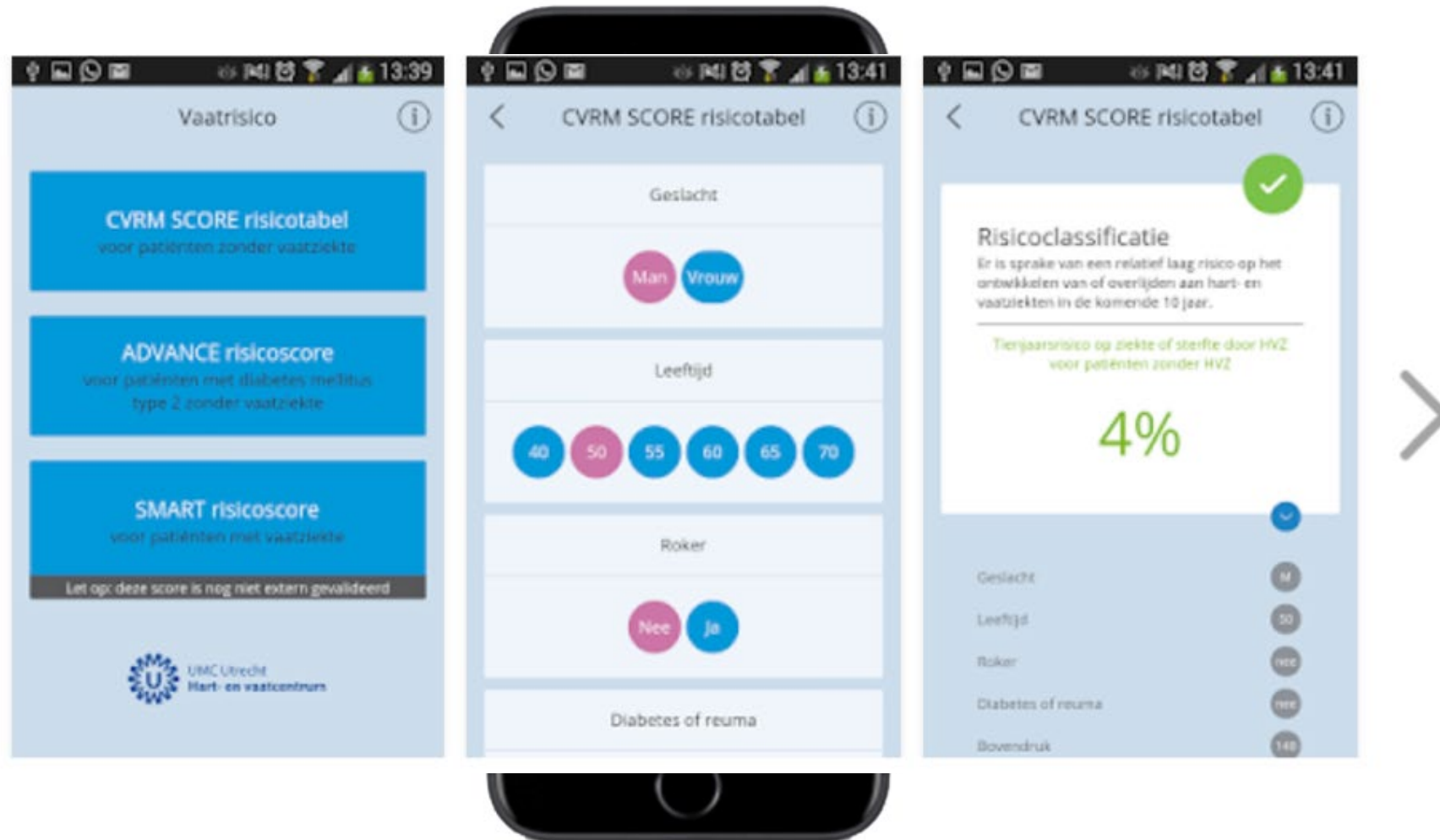


Gulshan et al. JAMA, 2016

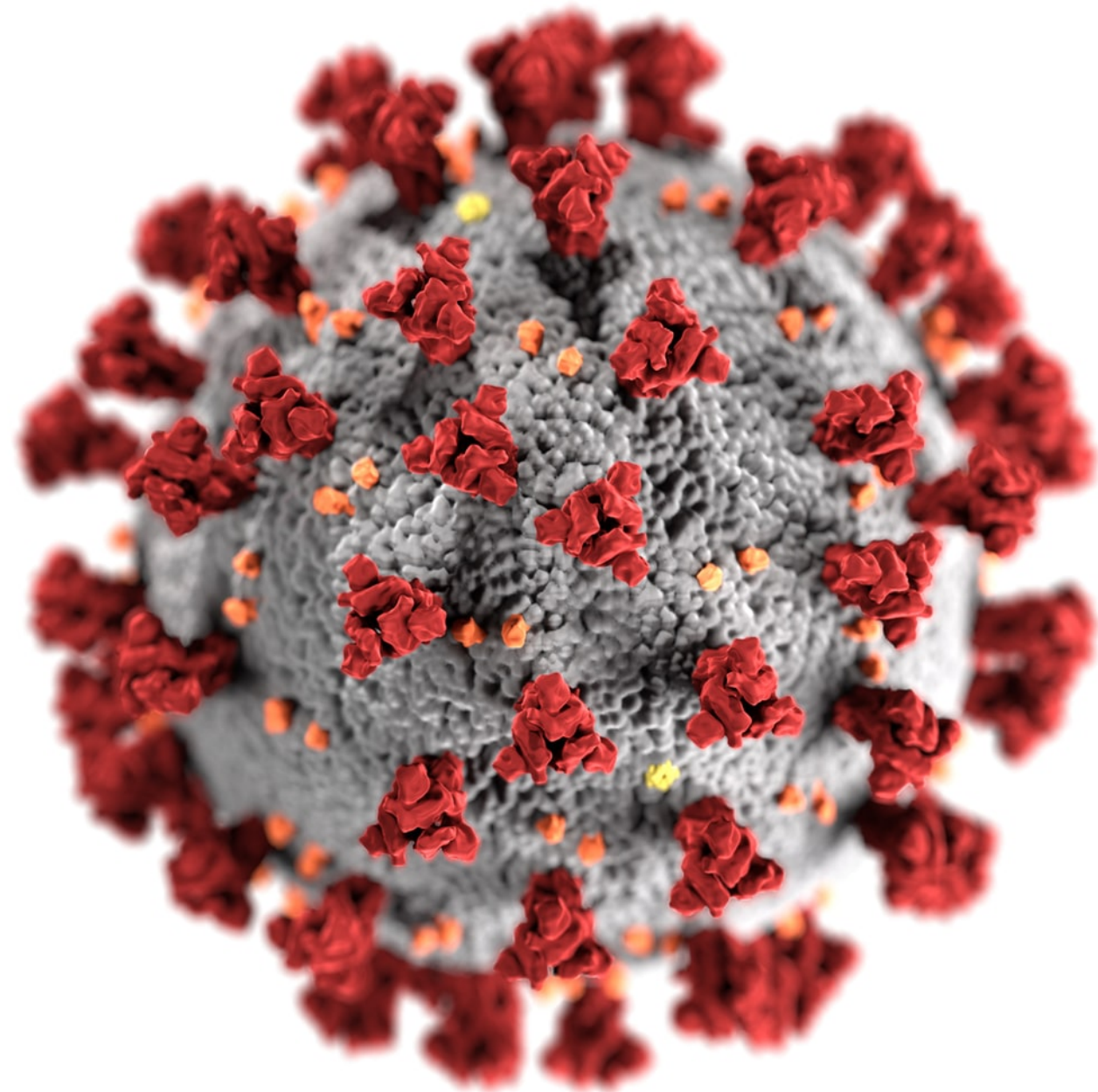
“Bedside” prediction models



“Bedside” prediction models



Numerical example: 10 year death due to cardiovascular disease





European Commission > Strategy > Shaping Europe's digital future > News >

Shaping Europe's digital future

NEWS ARTICLE | 19 May 2020

Using AI to fast and effectively diagnose COVID-19 in hospitals

The European Commission will invest in the use of Artificial Intelligence to speed up the diagnosis of COVID-19 and improve the future treatment of patients. A software developed to assist the work of medical staff by analysing images of pulmonary infections is introduced in 10 hospitals across Europe.

About Artificial intelligence

Policies

Blog posts

News

“As of today, we have deployed the system in 16 hospitals, and it is performing over 1,300 screenings per day”
MedRxiv pre-print only, 23 March 2020,
doi.org/10.1101/2020.03.19.20039354

Infervision's AI is in Italy Helping to Battle COVID-19

PRESS RELEASE UPDATED: MAR 23, 2020

ROME, March 20, 2020 (NewsWire.com) - COVID-19 is spreading, with European countries already declaring a pandemic. The World Health Organization has declared Europe as the new 'epicenter' for COVID-19. Italy announced a full lock-down on March 10. Due to the spreading of COVID-19, Italian medical institutions are facing tremendous pressure as patient numbers surge. Meanwhile, issues over long turnaround times for PCT testing and limited availabilities of the kit are concerning. Using CT images will help with the screening of COVID-19.



New tool could 'help UK doctors spot high-risk Covid patients in seconds'

Study claims risk calculator will help clinicians with expected influx of patients this autumn

- **Coronavirus - latest updates**
- **See all our coronavirus coverage**



▲ The calculator was tested in a hospitalised elderly population, so is not applicable for use within the community.
Photograph: Murdo MacLeod/The Guardian

score of three or less. The tool, which is easily accessible on a smartphone or computer, takes seconds to generate a score and is expected to be rolled out in the **NHS** this week.

Covid: Extra 1.7m vulnerable added to shielding list

By Nick Triggle
Health correspondent

🕒 1 day ago | 💬 Comments



There is to be a large expansion of the number of people being asked to shield in England.

An extra 1.7 million people are expected to be added to the 2.3 million already on the list.

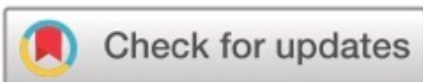
Half of the group have not yet been vaccinated so will now be prioritised urgently by their local GPs.

It comes after a **new model** was developed that takes into account extra factors rather than just health.

This calculation includes things such as ethnicity, deprivation (by postcode) and weight to work out a person's risk of becoming seriously ill if they were to catch Covid.

It also looks at age, underlying health issues and prescribed medications.

Prof Andrew Hayward, a member of the New and Emerging Respiratory Virus Threats Advisory Group (Nervtag), which has been involved in the modelling, said it considered a "combination of factors" such as age, ethnicity and chronic illness and put them together to reach a score.



FAST TRACK

Prediction models for diagnosis and prognosis of covid-19 infection: systematic review and critical appraisal

Laure Wynants,^{1,2} Ben Van Calster,^{2,3} Marc M J Bonten,^{4,5} Gary S Collins,^{6,7} Thomas P A Debray,^{4,8} Maarten De Vos,^{2,9} Maria C Haller,^{10,11} Georg Heinze,¹⁰ Karel G M Moons,^{4,8} Richard D Riley,¹² Ewoud Schuit,^{4,8} Luc J M Smits,¹ Kym I E Snell,¹² Ewout W Steyerberg,³ Christine Wallisch,^{10,13,14} Maarten van Smeden⁴

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Additional material is published online only. To view please visit the journal online.

Cite this as: *BMJ* 2020;369:m1328

ABSTRACT

OBJECTIVE

To review and critically appraise published and preprint reports of prediction models for diagnosing coronavirus disease 2019 (covid-19) in patients with suspected infection, for prognosis of patients with covid-19, and for detecting people in the general population at risk of being admitted to hospital for covid-19 pneumonia.

were identified for predicting hospital admission from pneumonia and other events (as proxy outcomes for covid-19 pneumonia) in the general population; 18 diagnostic models for detecting covid-19 infection (13 were machine learning based on computed tomography scans); and 10 prognostic models for predicting mortality risk, progression to severe disease, or length of hospital stay. Only one study used patient data from outside of China. The most



- Published on **7 April 2020**

- **18 days** between idea and article acceptance (*sprint*)

- Invited by BMJ as the first ever **living review** (*marathon*)

Prediction models for diagnosis and prognosis of covid-19: a systematic review and critical appraisal

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Cite this as: *BMJ* 2020;360:m1339

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Latest version (Update 3)

RESEARCH



OPEN ACCESS



FAST TRACK

Prediction models for diagnosis and prognosis of covid-19: systematic review and critical appraisal

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OBJECTIVE

To review and appraise the validity and usefulness of published and preprint reports of prediction models for diagnosing coronavirus disease 2019 (covid-19)

DATA SOURCES

PubMed and Embase through Ovid, up to 1 July 2020, supplemented with arXiv, medRxiv, and bioRxiv up to 5 May 2020.

STUDY SELECTION

Prediction model related to prognosis and diagnosis

3 main types models

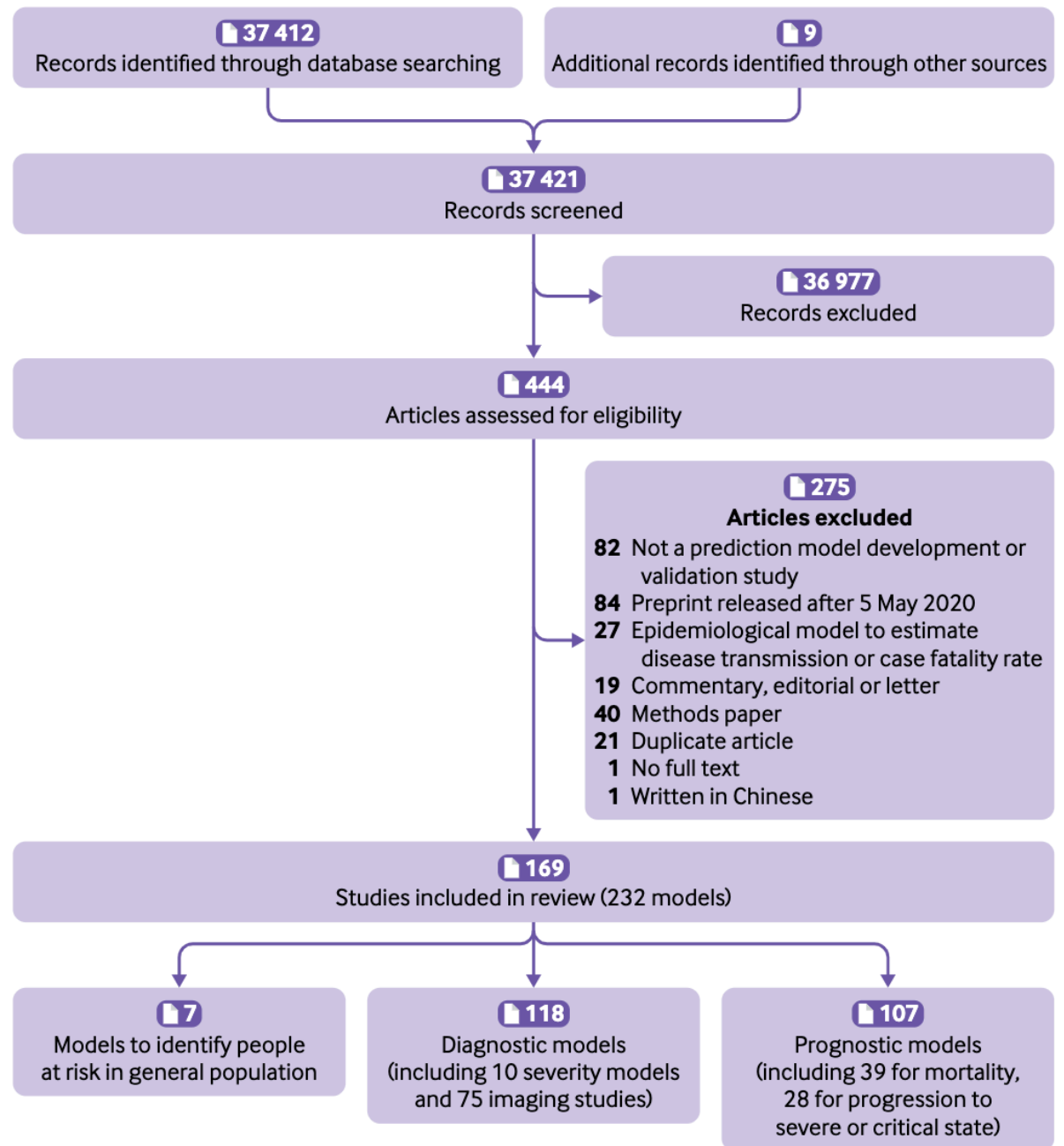
1. Patient X is infected / COVID-19 pneumonia
diagnostic
2. Patient X will die from COVID-19 / need respiratory support
prognostic
3. Currently healthy person X will get severe COVID-19
general population

Data extraction

- 43 researchers, duplicate reviews
- Extraction form based on **CHARMS** checklist & **PROBAST** (127 questions)
- Assessment of each prediction model separately if more than one was developed/validated

UPDATE 3

- **232 models** reviewed
- Peer reviewed articles
in Pubmed and Embase
- Pre-prints only until update 2
from bioRxiv, medRxiv, and arXiv
- Search: up to 1 July 2020



Some key numbers

Origin of data

- Single country: N = 174, **75%** (of which from China: 56%)
- Multiple countries: **18%**
- Unknown origin: **7%**

Target setting

- Patients admitted to hospital: N = 119, **51%**
- Patient at triage centre or fever clinic: **5%**
- Patients in general practice: **1%**
- Other/unclear: **42%**

Some key numbers

Target population

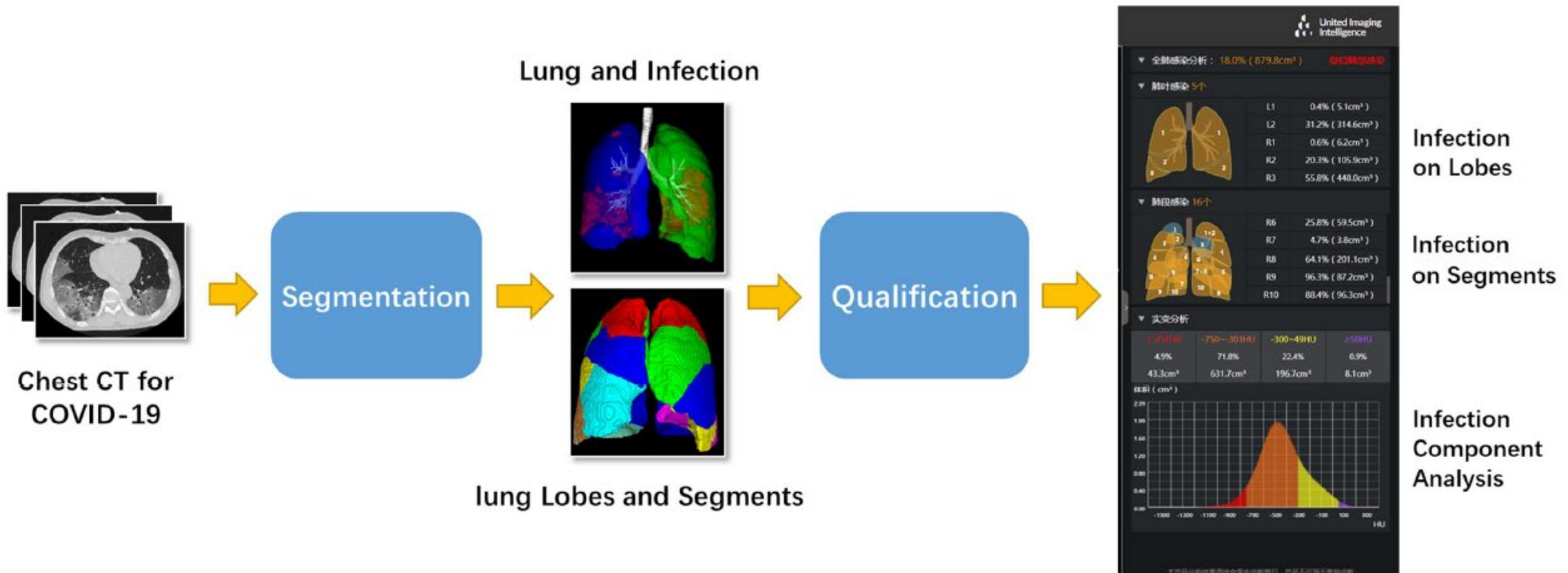
- Confirmed COVID-19: N = 108, **47%**
- Suspected COVID-19: **36%**
- Other/unclear: **18%**

Sample size

- Sample size development, median: **338 (IQR: 134—707)**
 - number of events, median: **69 (37—160)**
- Sample size external validation, median: **189 (76—312)**
 - Number of events, median: **40 (24—122)**

Models

- Logistic regression (including regularized): **35%**
- Neural net (including deep learning): **33%**
- Tree-based (including random forest): **7%**
- Survival models (including Cox ph): **6%**
- SVM: **4%**
- Other/unclear: **15%**

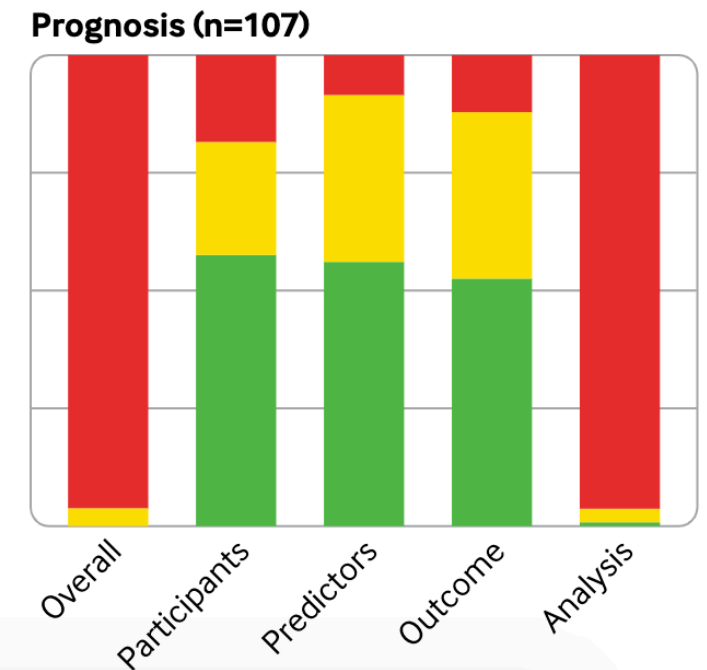
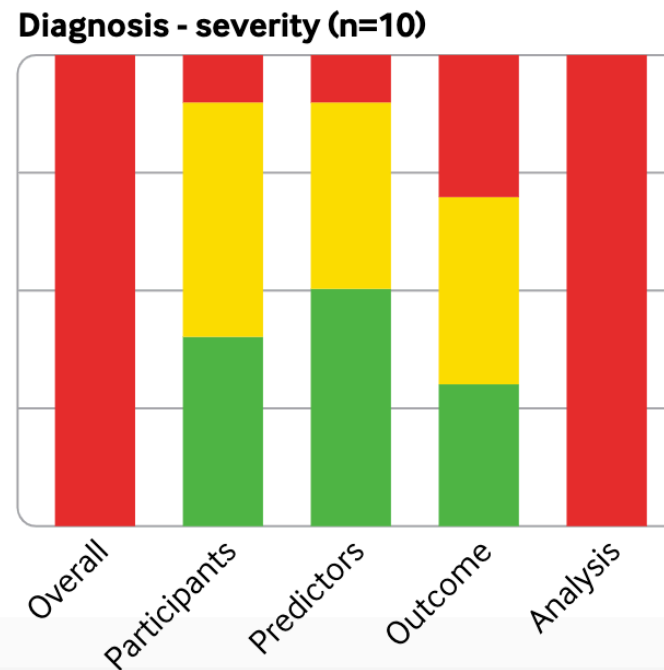
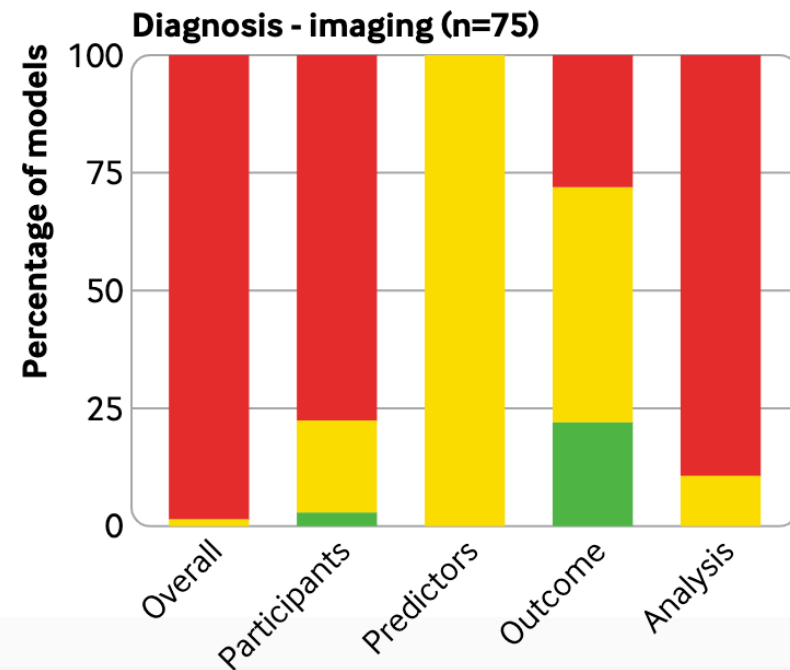
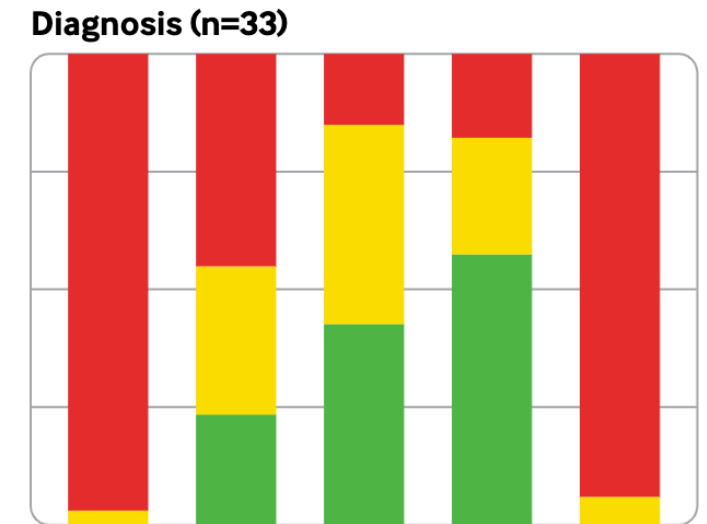
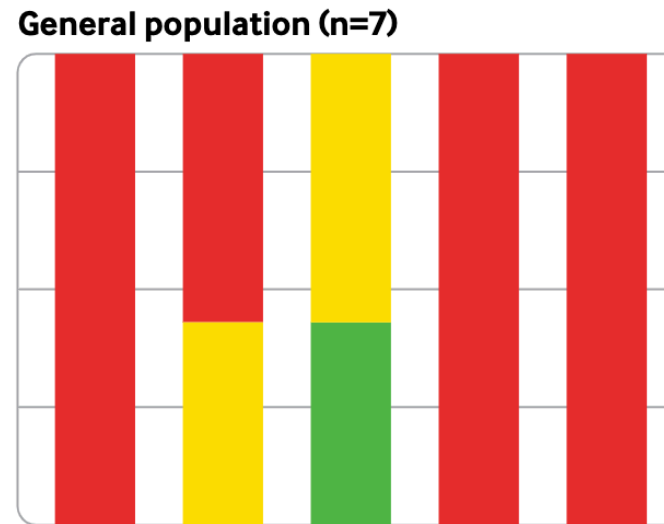
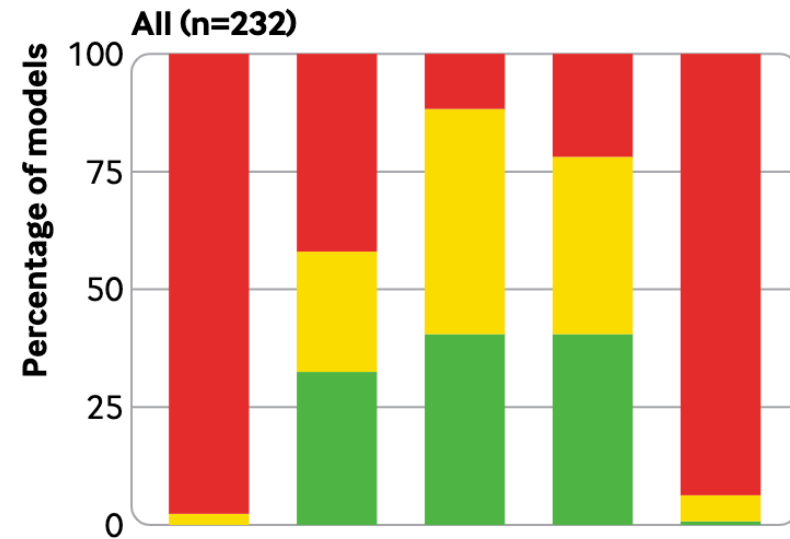


Reported performance – AUC range

- **General population models:** 0.71 to 0.99
- **Diagnostic models:** 0.65 to 0.99
- **Diagnostic severity models:** 0.80 to 0.99
- **Diagnostic imaging models:** 0.70 to 0.99
- **Prognosis models:** 0.54 to 0.99
prediction horizon varied from 1 to 37 days

Risk of bias

Low Unclear High



Risk of bias
Low Unclear High

Participants

- Inappropriate or unclear in/exclusion or study design

Predictors

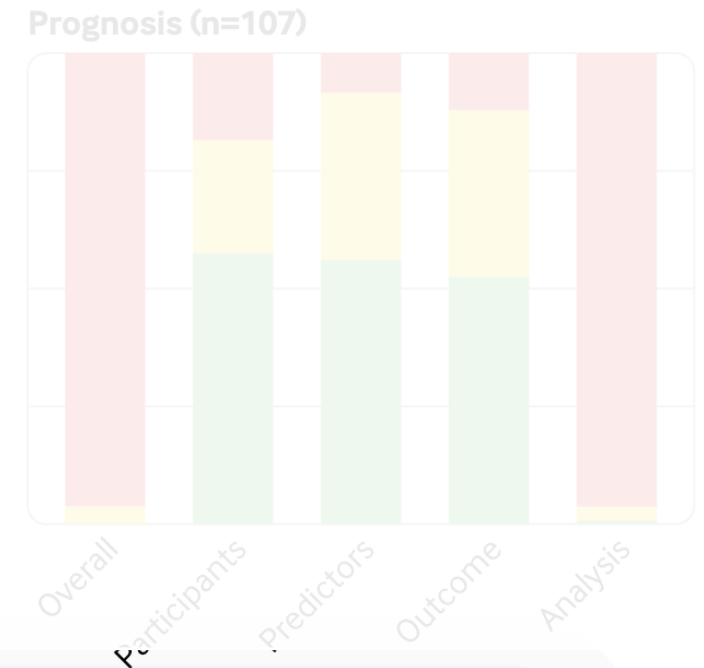
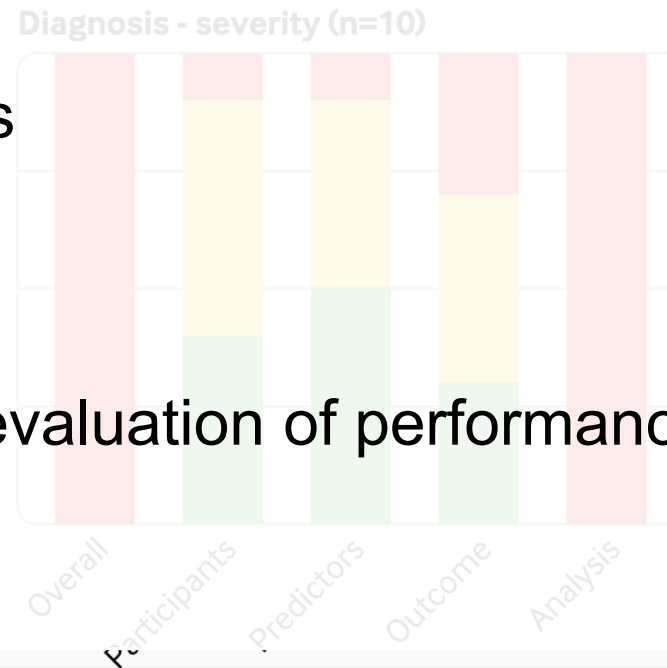
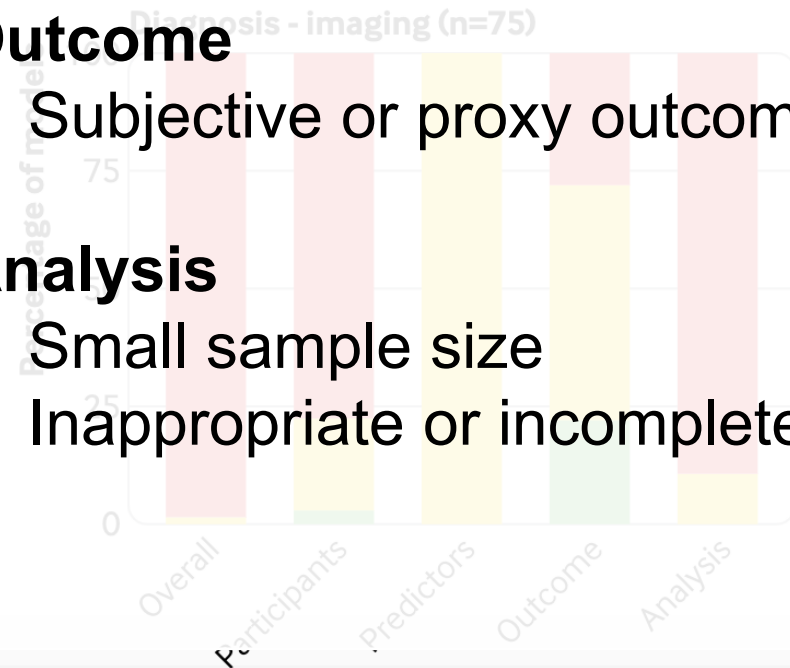
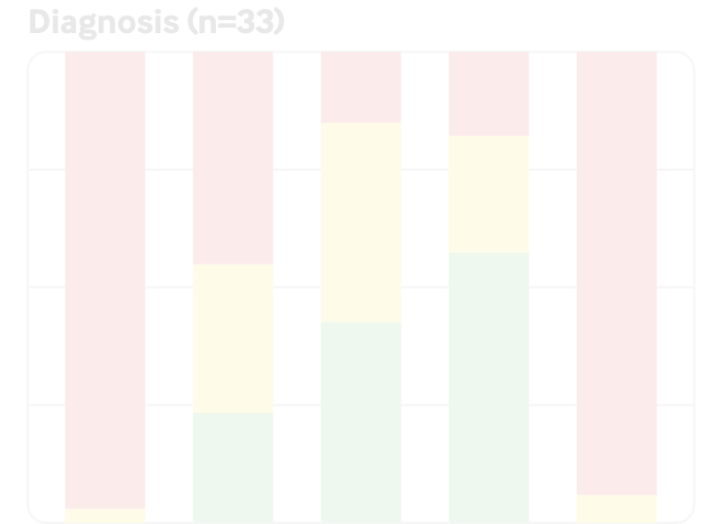
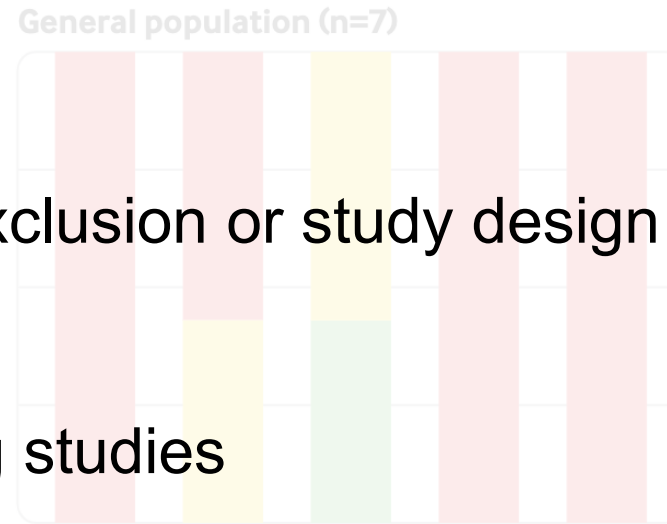
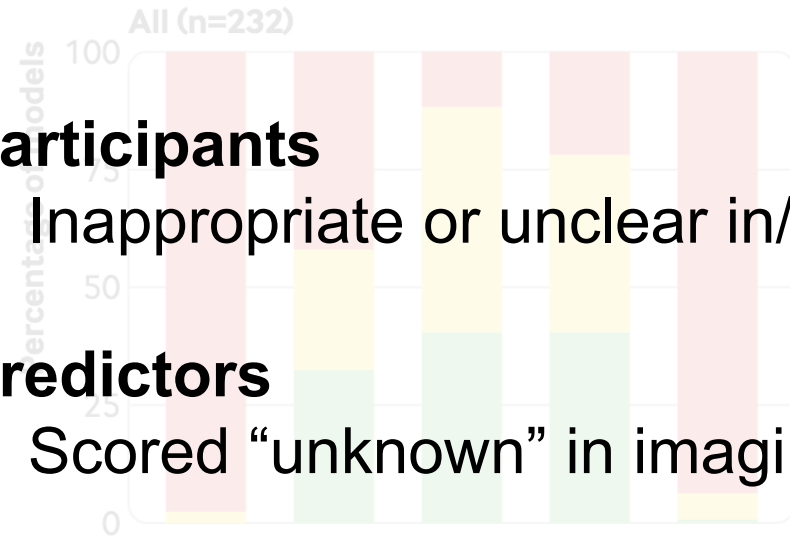
- Scored “unknown” in imaging studies

Outcome

- Subjective or proxy outcomes

Analysis

- Small sample size
- Inappropriate or incomplete evaluation of performance



Conclusion update 3 living review

“...models are all at high or unclear risk of bias”

We do “not recommend any of the current prediction models to be used in practice, but one diagnostic and one prognostic model originated from higher quality studies and should be (independently) validated in other datasets”



Eric Topol ✓
@EricTopol

There have been serious replication issues w/ #AI but now it has become prominent for #COVID19 applications. A recent mortality predictive model w/ 3 biomarkers (LDH, CRP, lymphocyte count) was shown to be irreproducible by 3 groups @NatMachIntell

nature machine intelligence
ARTICLES
<https://doi.org/10.1038/s42256-020-01862-7>

An interpretable mortality prediction model for COVID-19 patients

Li Yan^{1,2}, Hai-Tao Zhang^{1,2}, Jorgo Goncalves^{1,2,3}, Yang Xiao², Maslin Wang², Yuesi Guo², Chuan Sun¹, Xiuchuan Tang¹, Liang Jing¹, Mingyang Zhang¹, Xiang Huang², Ying Xiao¹, Haosen Cao¹, Yanyan Chen¹, Tongxin Ren¹, Fang Wang¹, Yaru Xiao¹, Sufang Huang¹, Xi Tan¹, Mianmian Huang¹, Bo Jiao¹, Chang Cheng¹, Yong Zhang¹, Ailin Luo¹, Laurent Mombaerts¹, Junyang Jin¹, Zhiguo Cao¹, Shusheng Li^{1,2}, Hai Xu^{1,2} and Ye Yuan^{1,2}

The sudden increase in COVID-19 cases is putting high pressure on healthcare services worldwide. At this stage, fast, accurate and early clinical assessment of the disease severity is vital. To support decision making and logistical planning in healthcare systems, this study leverages a database of blood samples from 483 infected patients in the region of Wuhan, China, to identify crucial predictive biomarkers of disease mortality. For this purpose, machine learning tools selected those biomarkers that enabled the mortality of individual patients more than 30 days in advance with more than 90% accuracy in the independent validation.

nature machine intelligence
MATTERS ARISING
<https://doi.org/10.1038/s42256-020-00204-2>

External validation demonstrates limited clinical utility of the interpretable mortality prediction model for patients with COVID-19

Matthew Barish^{1,2,3}, Sivash Bolourani^{1,2,3,4,5}, Lawrence F. Lau^{1,2,3}, Sareen Shih^{1,2,3} and Theodoros P. Zanos^{1,2,3,4}

Typical artificial intelligence or machine learning algorithm deployment begins with model training followed by testing under varying circumstances to enable model adaptation and verification of a model's performance. Despite the urgency that the coronavirus

on their validation data at the first time point that all three blood tests were collected, similar to how physicians would risk-stratify new patients (Fig. 1c). This critical time point was not included in the original paper, which is important as the authors suggest that

nature machine intelligence
MATTERS ARISING
<https://doi.org/10.1038/s42256-020-00204-2>

Replication of a mortality prediction model in Dutch patients with COVID-19

Marian J. R. Quenel^{1,2}, Thijs C. van Hulten¹, Pietermel C. Garst-van der Vliet¹, Jette Wieland¹, Bekir Karakaya¹, Maaike Söhne¹, Hazro S. Moeniralam¹ and Jan C. Grutters^{1,2}

There is a clear need for a simple mortality prediction model to help guide clinical decision-making for patients with COVID-19. Yan et al. demonstrated the strong predictive capacity of a decision rule consisting of three readily available laboratory measures for COVID-19 mortality: lactate dehydrogenase (LDH), high-sensitivity C-reactive protein (hs-CRP) and percent lymphocytes. We performed an independent replication of their model.

nature machine intelligence
MATTERS ARISING
<https://doi.org/10.1038/s42256-020-00204-2>

Limited applicability of a COVID-19 specific mortality prediction rule to the intensive care setting

C. Dupuis^{1,2}, E. De Montrollé^{1,2}, M. Nouville^{1,2}, B. Moaville¹, S. Ruckly¹ and J. F. Tiraja^{1,2}

We read the Article by Yan et al. with great interest. The COVID-19 pandemic has established itself as a major burden on healthcare systems worldwide. Several algorithms to optimize the use of healthcare resources are of paramount importance. Against this background, Yan et al. gathered samples from a cohort of 483 infected patients in the region of Wuhan, China with a high mortality rate (30.2%) and proposed a mortality prediction model.

Characteristic	No./118 patients
Age	47 (32–69)
Sex (Male/F)	103 (87.3%)

<https://twitter.com/EricTopol/status/1328086389024440321>



OPEN

Common pitfalls and recommendations for using machine learning to detect and prognosticate for COVID-19 using chest radiographs and CT scans

Michael Roberts^{1,2}✉, Derek Driggs¹, Matthew Thorpe³, Julian Gilbey¹, Michael Yeung⁴,
Stephan Ursprung^{4,5}, Angelica I. Aviles-Rivero¹, Christian Etmann¹, Cathal McCague^{4,5},
Lucian Beer⁴, Jonathan R. Weir-McCall^{4,6}, Zhongzhao Teng⁴, Effrossyni Gkrania-Klotsas⁷,
AIX-COVNET*, James H. F. Rudd^{8,36}, Evis Sala^{4,5,36} and Carola-Bibiane Schönlieb^{1,36}

Machine learning methods offer great promise for fast and accurate detection and prognostication of coronavirus disease 2019 (COVID-19) from standard-of-care chest radiographs (CXR) and chest computed tomography (CT) images. Many articles have been published in 2020 describing new machine learning-based models for both of these tasks, but it is unclear which are of potential clinical utility. In this systematic review, we consider all published papers and preprints, for the period from 1 January 2020 to 3 October 2020, which describe new machine learning models for the diagnosis or prognosis of COVID-19 from CXR or CT images. All manuscripts uploaded to bioRxiv, medRxiv and arXiv along with all entries in EMBASE and MEDLINE in this timeframe are considered. Our search identified 2,212 studies, of which 415 were included after initial screening and, after quality screening, 62 studies were included in this systematic review. **Our review finds that none of the models identified are of potential clinical use due to methodological flaws and/or underlying biases.** This is a major weakness, given the urgency with which validated COVID-19 models are needed. To address this, we give many recommendations which, if followed, will solve these issues and lead to higher-quality model development and well-documented manuscripts.

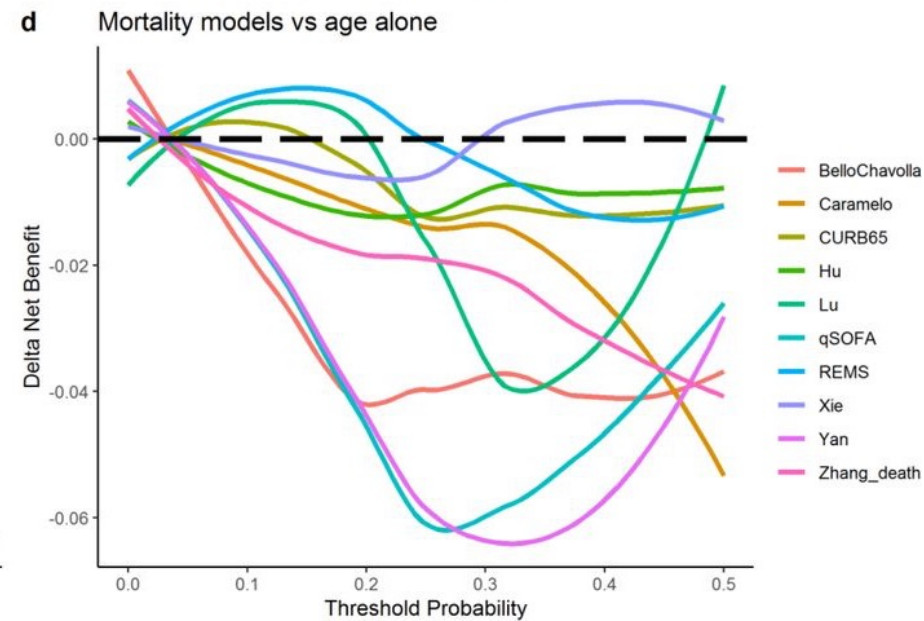
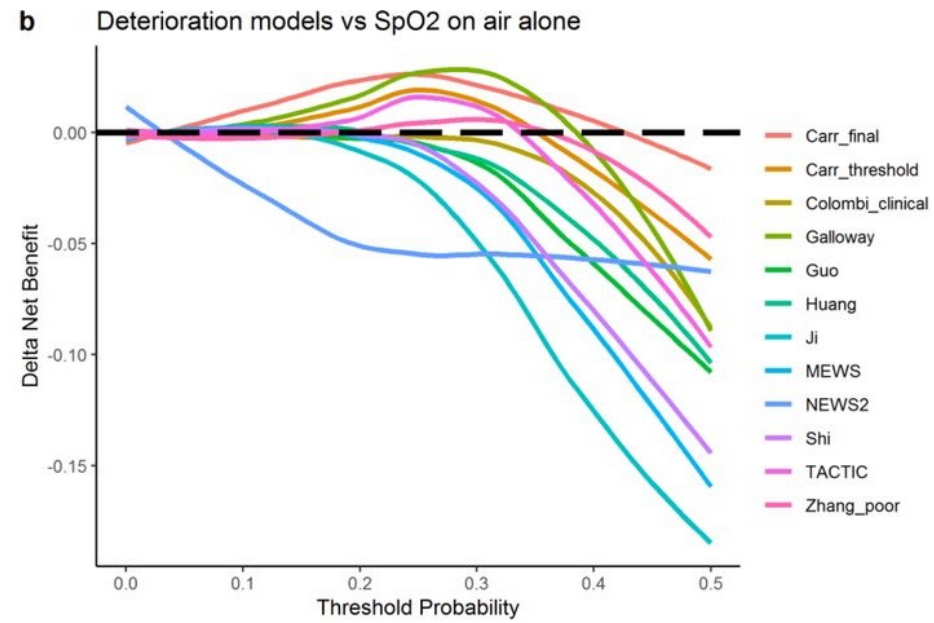
Systematic evaluation and external validation of 22 prognostic models among hospitalised adults with COVID-19: an observational cohort study

Rishi K. Gupta ^{1,2}, Michael Marks ^{2,3}, Thomas H.A. Samuels², Akish Luintel², Tommy Rampling², Humayra Chowdhury², Matteo Quartagno⁴, Arjun Nair², Marc Lipman ⁵, Ibrahim Abubakar ¹, Maarten van Smeden ⁶, Wai Keong Wong², Bryan Williams^{7,8} and Mahdad Noursadeghi ^{2,9}, on behalf of The UCLH COVID-19 Reporting Group¹⁰



@ERSpublications

Oxygen saturation on room air and patient age are strong predictors of deterioration and mortality, respectively, among hospitalised adults with COVID-19. None of the 22 prognostic models evaluated in this study adds incremental value to these univariable predictors. <https://bit.ly/2Hg24TO>

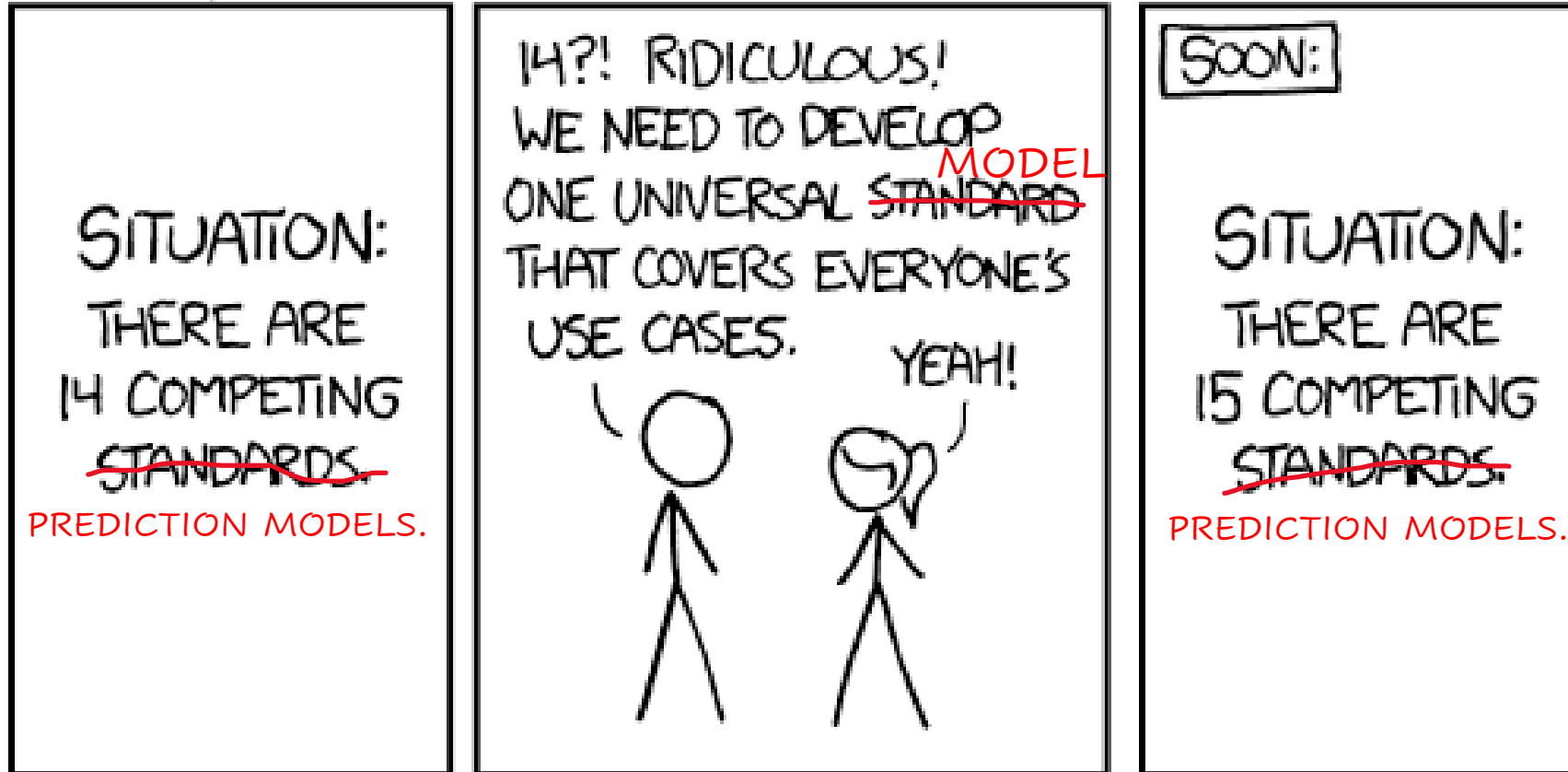


Gupta, ERJ, 2020

Landscape of clinical prediction models

- 408 models for COPD prognosis (Bellou, 2019)
- 363 models for cardiovascular disease general population (Damen, 2016)
- 263 prognosis models in obstetrics (Kleinrouweler, 2016)
- 258 models mortality after general trauma (Munter, 2017)
- 232 models related to COVID-19 (Wynants, 2020)
- 160 female-specific models for cardiovascular disease (Baart, 2019)
- 119 models for critical care prognosis in LMIC (Haniffa, 2018)
- 101 models for primary gastric cancer prognosis (Feng, 2019)
- 81 models for sudden cardiac arrest (Carrick, 2020)
- 74 models for contrast-induced acute kidney injury (Allen, 2017)
- 73 models for 28/30 day hospital readmission (Zhou, 2016)
- 68 models for preeclampsia (De Kat, 2019)
- 67 models for traumatic brain injury prognosis (Dijkland, 2019)
- 64 models for suicide / suicide attempt (Belsher, 2019)
- 61 models for dementia (Hou, 2019)
- 58 models for breast cancer prognosis (Phung, 2019)
- 52 models for pre-eclampsia (Townsend, 2019)
- 52 models for colorectal cancer risk (Usher-Smith, 2016)
- 48 models for incident hypertension (Sun, 2017)
- 46 models for melanoma (Kaiser, 2020)
- 46 models for prognosis after carotid revascularisation (Volkers, 2017)
- 43 models for mortality in critically ill (Keuning, 2019)
- 42 models for kidney failure in chronic kidney disease (Ramspek, 2019)
- 40 models for incident heart failure (Sahle, 2017)
- 37 models for treatment response in pulmonary TB (Peetluk, 2021)
- 35 models for in vitro fertilisation (Ratna, 2020)
- 34 models for stroke in type-2 diabetes (Chowdhury, 2019)
- 34 models for graft failure in kidney transplantation (Kabore, 2017)
- 31 models for length of stay in ICU (Verburg, 2016)
- 27 models for pediatric early warning systems (Trubey, 2019)
- 27 models for malaria prognosis (Njim, 2019)
- 26 models for postoperative outcomes colorectal cancer (Souwer, 2020)
- 26 models for childhood asthma (Kothalawa, 2020)
- 25 models for lung cancer risk (Gray, 2016)
- 25 models for re-admission after admitted for heart failure (Mahajan, 2018)
- 23 models for recovery after ischemic stroke (Jampathong, 2018)
- 23 models for delirium in older adults (Lindroth, 2018)
- 21 models for atrial fibrillation detection in community (Himmelreich, 2020)
- 19 models for survival after resectable pancreatic cancer (Stijker, 2019)
- 18 models for recurrence hep. carcinoma after liver transplantation (Al-Ameri, 2020)
- 18 models for future hypertension in children (Hamoen, 2018)
- 18 models for risk of falls after stroke (Walsh, 2016)
- 18 models for mortality in acute pancreatitis (Di, 2016)
- 17 models for bacterial meningitis (van Zeggeren, 2019)
- 17 models for cardiovascular disease in hypertensive population (Cai, 2020)
- 14 models for ICU delirium risk (Chen, 2020)
- 14 models for diabetic retinopathy progression (Haider, 2019)

PREDICTION MODELS
HOW ~~STANDARDS~~ PROLIFERATE:
(SEE: A/C CHARGERS, CHARACTER ENCODINGS, INSTANT MESSAGING, ETC.)

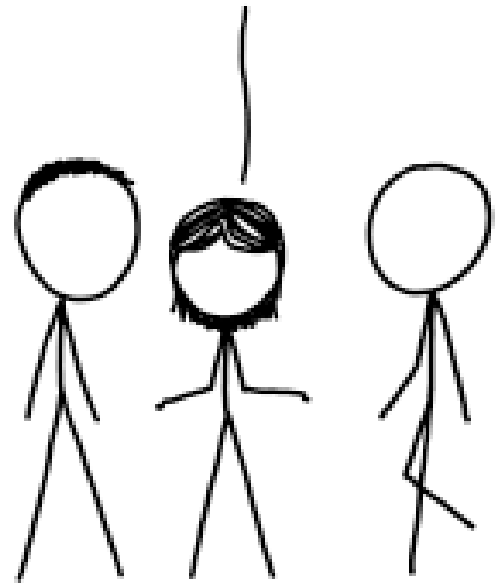


Original: <https://xkcd.com/927/>

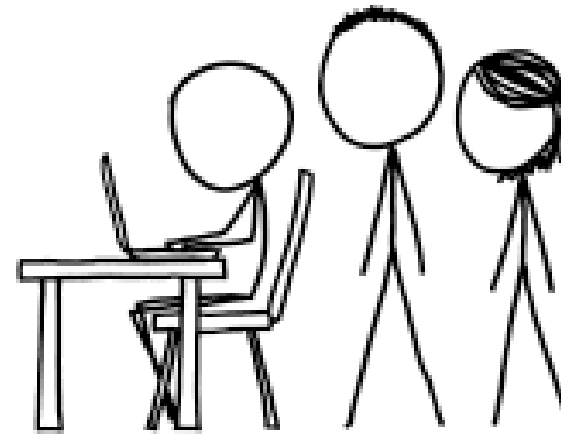
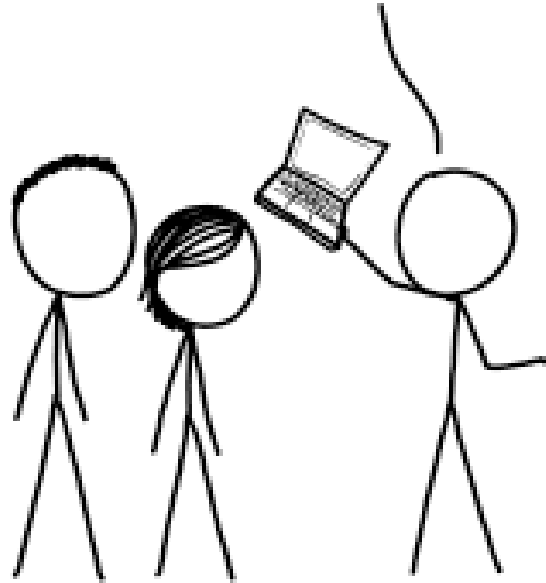
Agenda

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 - Applied example: ADNEX model
 - Future perspective: machine learning and AI

OUR FIELD HAS BEEN
STRUGGLING WITH THIS
PROBLEM FOR YEARS.



STRUGGLE NO MORE!
I'M HERE TO SOLVE
IT WITH *ALGORITHMS!*



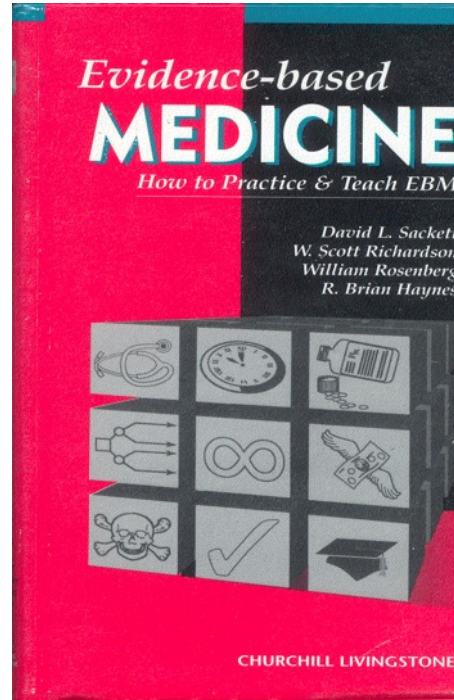
SIX MONTHS LATER:

WOW, THIS PROBLEM
IS REALLY HARD.

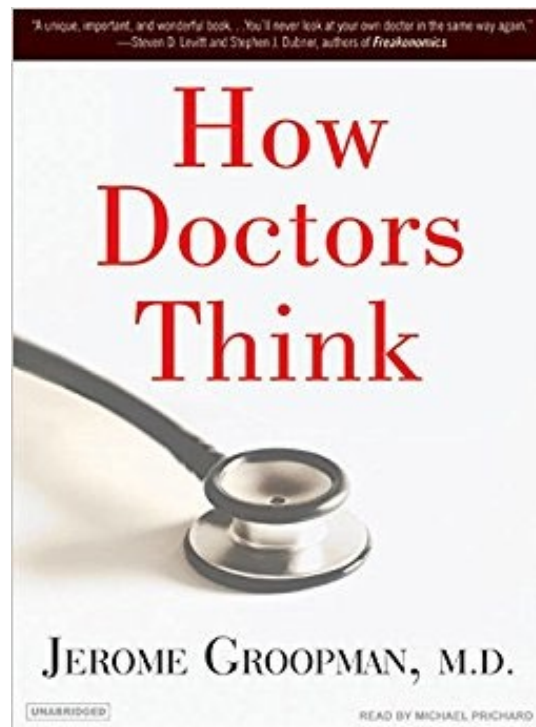
YOU DON'T SAY.



Diagnosis: “*starting point of all medical actions*”



*"Usually doctors are right, but conservatively about **15 percent** of all people are **misdiagnosed**.
Some experts think it's as high as **20 to 25 percent**"*



*“Patient trust was essential in the healing process. It could be won by a punctilious bedside manner, by meticulous explanation, and by **mastery of prognosis**, an art of demanding experience, observation and logic”*

Galen, 2nd century AD

PROGRESS framework

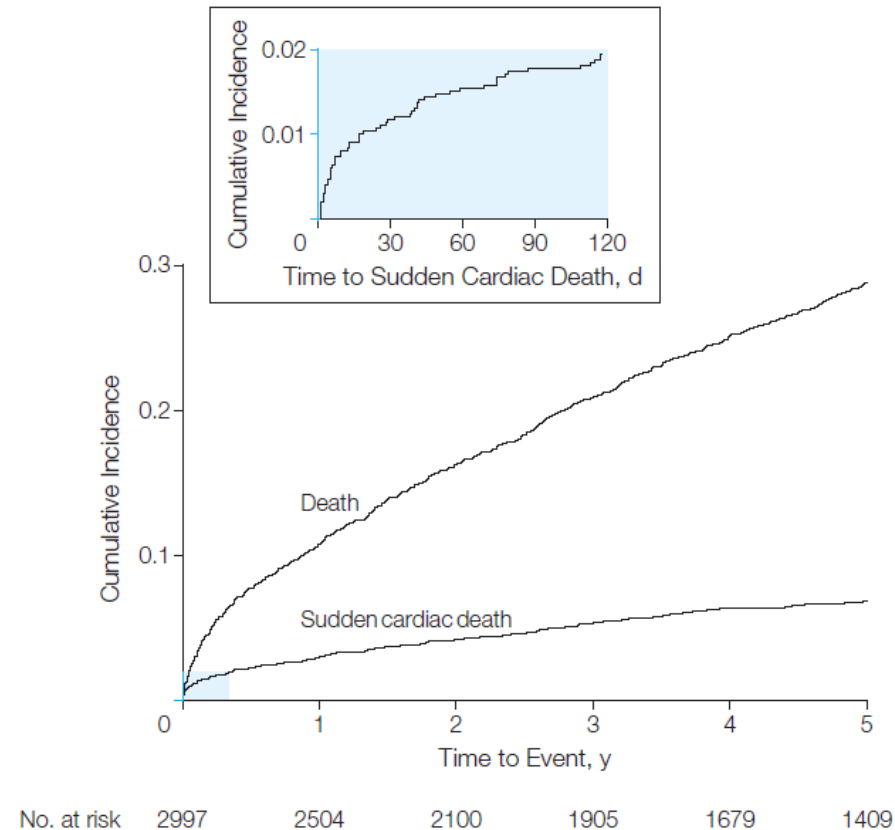
The PROGRESS (PROGnosis RESearch Strategy) framework classifies prognosis research into four main types of study:

1. Overall prognosis research
2. Prognostic factor research
3. Prognostic model research
4. Predictors of treatment effect research

More details: <https://www.prognosisresearch.com/progress-framework>

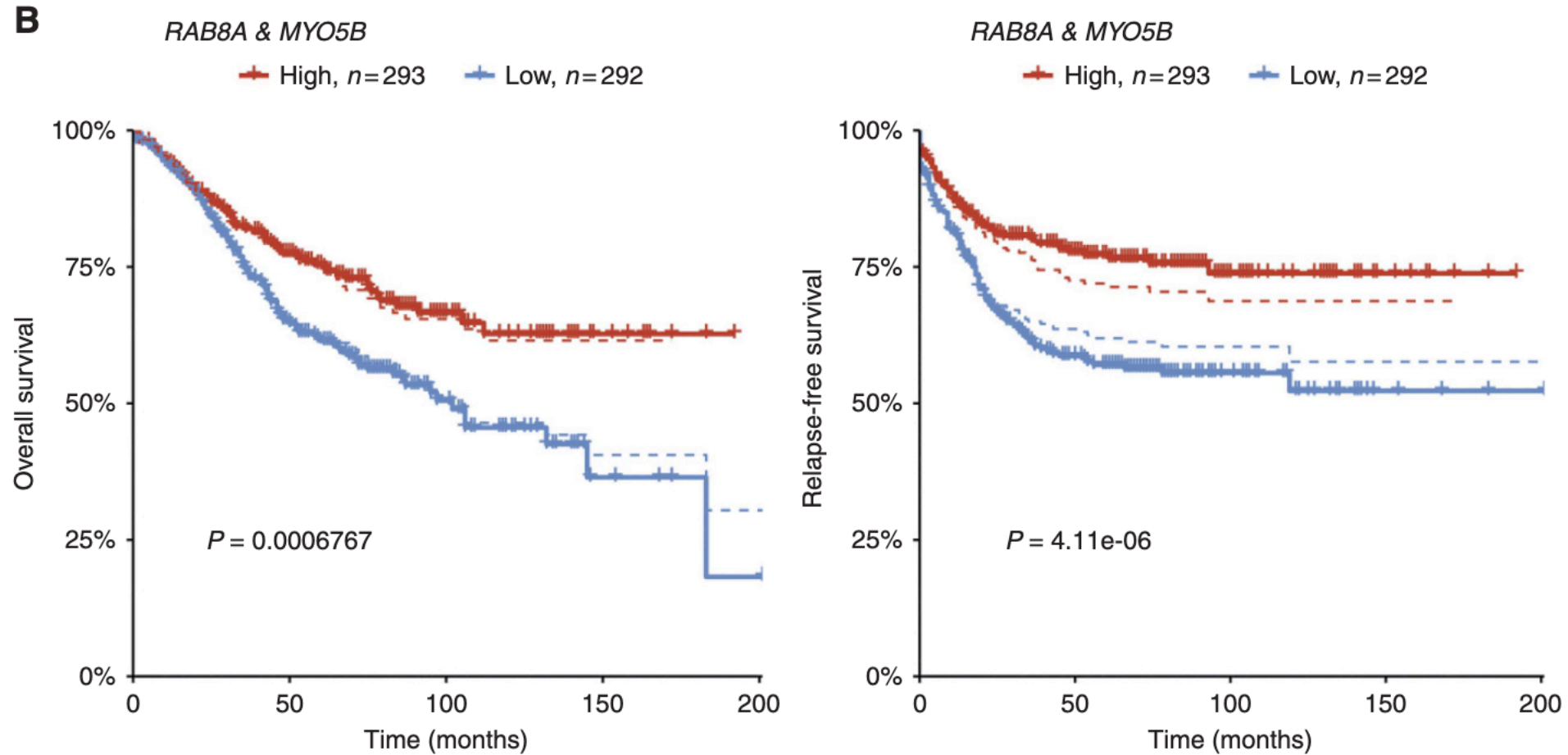
Overall prognosis research

Figure 1. Cumulative Incidence of Sudden Cardiac Death and All-Cause Mortality After Myocardial Infarction Among Residents of Olmsted County, Minnesota



The blue area on the plot represents the cumulative incidence of sudden death during the first 120 days after the index myocardial infarction.

Prognostic factor research



Letellier et al. BJC, 2017

Predictors of treatment effect research

ORIGINAL ARTICLE

Endocrine Care

Biomarkers as Predictors of Response to Treatment with Motesanib in Patients with Progressive Advanced Thyroid Cancer

Michael B. Bass, Steven I. Sherman, Martn J. Schlumberger, Michael T. Davis, Lisa Kivman, Huan-Mei Khoo, Kimberly H. Notari, Matthew Peach, Yong-jiang Hei, and Scott D. Patterson

Conclusions: Changes in PIGF and soluble VEGF receptor 2 levels after initiation of therapy predicted response to motesanib in patients with advanced differentiated thyroid cancer or metastatic medullary thyroid cancer. Lower baseline VEGF levels were associated with longer PFS. (*J Clin Endocrinol Metab* 95: 5018–5027, 2010)

Bass et al. JCEM, 2010

PROGRESS framework

The PROGRESS (PROGnosis RESearch Strategy) framework classifies prognosis research into four main types of study:

- | | |
|--|------------------------------------|
| 1. Overall prognosis research | Prevalence studies |
| 2. Prognostic factor research | Diagnostic test (accuracy) studies |
| 3. Prognostic model research | Diagnostic model research |
| 4. Predictors of treatment effect research | |

More details: <https://www.prognosisresearch.com/progress-framework>

Focus of this lecture

The PROGRESS (PROGnosis RESearch Strategy) framework classifies prognosis research into four main types of study:

1. Overall prognosis research Prevalence studies
2. Prognostic factor research Diagnostic test (accuracy) studies
- 3. Prognostic model research Diagnostic model research**
4. Predictors of treatment effect research

More details: <https://www.prognosisresearch.com/progress-framework>

The phases of prognostic/diagnostic model research

1. Development (including internal validation)
 2. External validation(s) and updating
 3. Software development and regulations
 4. Impact assessment
 5. Clinical implementation and scalability
- Most research focusses on development
 - Few models are validated
 - Fewer models reach the implementation



Not fit for purpose

- Wrong target population
- Expensive predictors
- Discrepancies between development and use
- Complexity/transparency

No validation/impact

- Poor development
- Insufficient reporting
- Incentives
- If done, usually small studies

Regulation/implementation

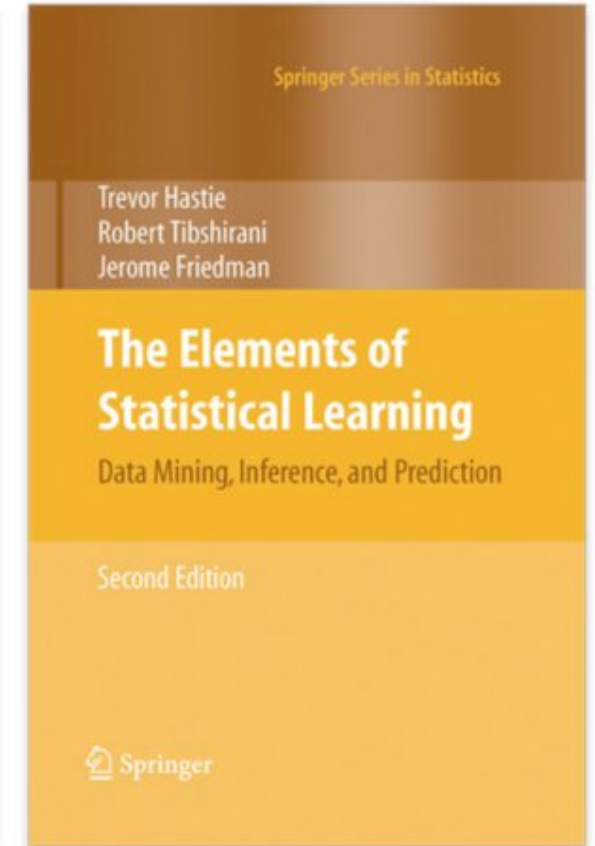
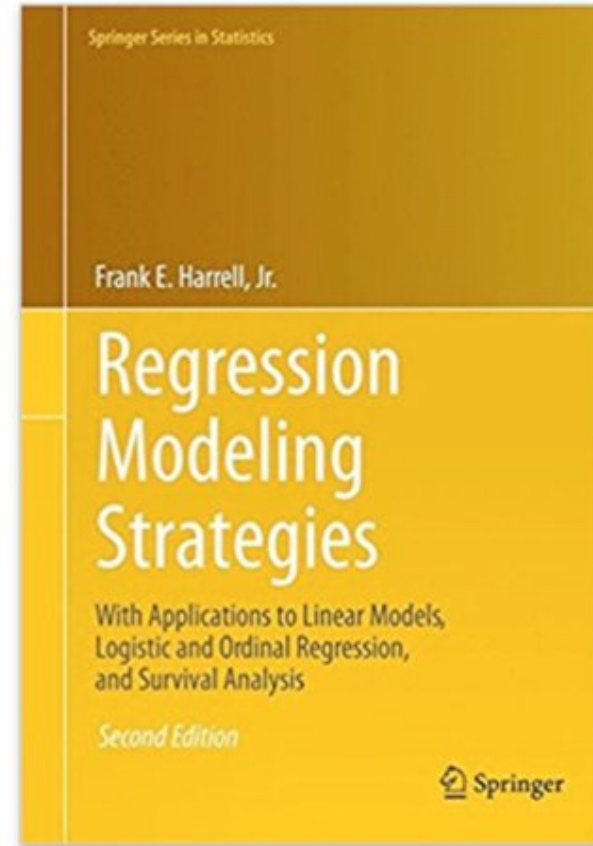
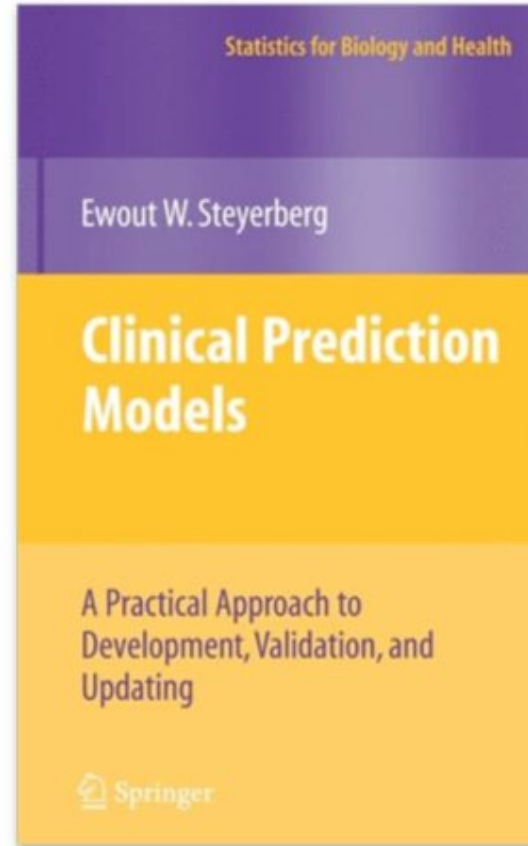
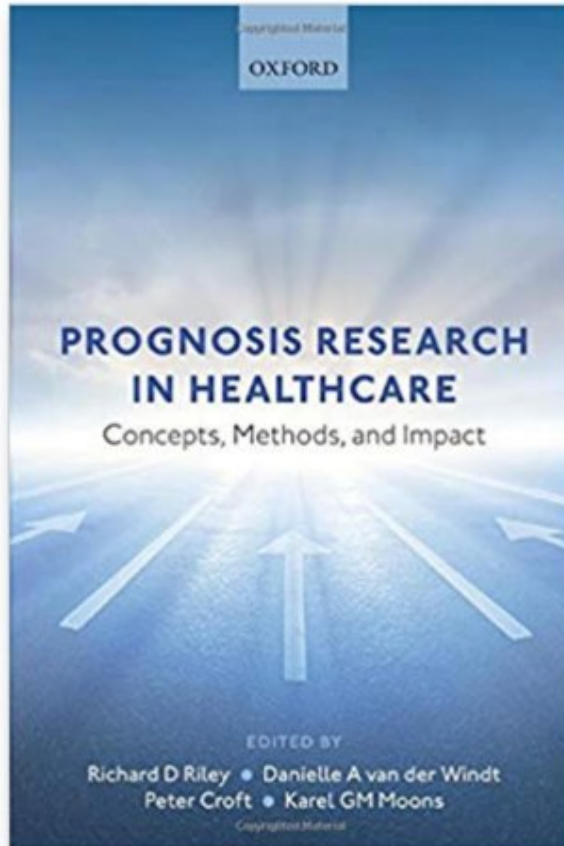
- MDR
- Soft- and hardware
- Model updating
- Quality control

Not adopted

- User time
- No prediction needed

Picture courtesy: Laure Wynants

Books (focus on development/validation)



Typical regression models

Binary logistic model

$$\begin{aligned}\Pr(Y = 1) &= \text{expit}(\beta_0 + \beta_1 X_1 + \dots + \beta_P X_P) \\ &= \exp(X\beta) / [1 + \exp(X\beta)]\end{aligned}$$

Multinomial logistic model

$$\Pr(Y = j) = \exp(X\beta_j) / [1 + \sum_{h=1}^{J-1} \exp(X\beta_h)]$$

Cox model

$$\begin{aligned}h(X, t) &= h_0(t) \exp(\beta_1 X_1 + \dots + \beta_P X_P) \\ &= h_0(t) \exp(X\beta)\end{aligned}$$

Remarks

- $X\beta$ is the linear predictor
- Most predictive performance metrics do not directly generalize between outcomes (discrimination/calibration)
- Linearity/additivity assumptions can be relaxed
- Tree based models (e.g. random forest), SVM and neural networks seem to be upcoming

Model users often unaware of models and equations

Prediction models for 6 month outcome after TBI

Admission Characteristics	Value
<i>Core</i>	
Age (14-99 years)	34
Motor Score	Normal Flexion
Pupils	Both reacting
<i>Core+CT</i>	
Hypoxia	No
Hypotension	No
CT Classification	Diffuse Injury III
tSAH on CT	Yes
Epidural mass on CT	Yes
<i>Core+CT+Lab</i>	
Glucose (3-20 mmol/L)	10 mmol/L
Hb (6-17 g/dL)	10 g/dL
Calculate	Reset

This model predicts outcome in the following patients:
Adults with head injury, Glasgow Coma Scale 12 or less.

Prognostic Results:

Predicted probability of 6 month mortality: **Core model: 17%**
Predicted probability of 6 month unfavourable outcome: **Core model: 32%**

Predicted probability of 6 month mortality: **Core+CT model: 20%**
Predicted probability of 6 month unfavourable outcome: **Core+CT model: 31%**

<http://www.tbi-impact.org/?p=impact/calc>

Reporting

Table 3 Prognostic factors for survival of patients with metastatic breast cancer (multivariate analysis)

Prognostic factors/index	<i>p</i>	Hazard ratio	95% CI
KPS	0.030	0.563	0.336–0.945
Nottingham prognostic groups	<0.0001	5.147	3.272–8.099
Liver metastasis	0.275	1.367	0.780–2.394
Number of metastasis	0.190	1.418	0.841–2.391
First-/second-/third-line chemotherapy for recurrence or metastasis	0.002	0.503	0.324–0.780

Baseline hazard?

Example: Systematic review by Bellou et al, 2019

Prognostic models for COPD: 228 articles
(**408 prognostic models**, 38 external validations)

Not reported:

- 12%: **modelling method** (e.g. type of regression model)
- 24%: evaluation of **discrimination** (e.g. area under the ROC curve)
- 64%: **missing data** handling (e.g. multiple imputation)
- 70%: **full model** (e.g. no regression coefficients, intercept / baseline hazard)
- 78%: no evaluation of **calibration** (e.g. calibration plot)

Reporting guidelines & risk of bias tools

- **TRIPOD**: reporting development/validation prediction models
- **PROBAST**: risk of bias development/validation prediction models
- **STARD**: reporting diagnostic accuracy studies
- **QUADAS-2**: risk of bias diagnostic accuracy studies
- **REMARK**: reporting (tumour marker) prognostic factor studies
- **QUIPS**: risk of bias in prognostic factor studies
- Currently in development: TRIPOD-AI, TRIPOD-cluster, PROBAST-AI, STARD-AI, QUADAS-AI, DECIDE-AI etc.

More info: equator-network.org

Dichotomania

*“**Dichotomania** is an obsessive compulsive disorder to which medical advisors in particular are prone... Show a medical advisor some continuous measurements and he or she immediately wonders. ‘Hmm, how can I make these clinically meaningful? Where can I cut them in two? What ludicrous side conditions can I impose on this?’”*

Stephen Senn

It's all in the title

1. **Problems in dichotomizing continuous variables** (Altman 1994)
2. **Dangers of using "optimal" cutpoints** in the evaluation of prognostic factors. (Altman et al 1994)
3. **How bad is categorization?** (Weinberg; 1995)
4. **Seven reasons why you should NOT categorize continuous data** (Dinero; 1996)
5. Breaking Up is Hard to Do: The **Heartbreak of Dichotomizing Continuous Data** (Streiner; 2002)
6. **Negative consequences of dichotomizing continuous predictor variables** (Irwin & McClelland; 2003)
7. **Why carve up your continuous data?** (Owen 2005)
8. **Chopped liver? OK. Chopped data? Not OK. Chopped liver? OK. Chopped data? Not OK** (Butts & Ng 2005)
9. **Categorizing continuous variables resulted in different predictors** in a prognostic model for nonspecific neck pain (Schellingerhout et al 2006)
10. **Dichotomizing continuous predictors in multiple regression: a bad idea** (Royston et al 2006)
11. The **cost of dichotomising continuous variables** (Altman & Royston; 2006)
12. **Leave 'em alone - why continuous variables should be analyzed as such** (van Walraven & Hart; 2008)
13. **Dichotomization of continuous data--a pitfall** in prognostic factor studies (Metze; 2008)
14. Analysis by **categorizing or dichotomizing continuous variables is inadvisable**: an example from the natural history of unruptured aneurysms (Naggara et al 2011)
15. **Against quantiles: categorization of continuous variables** in epidemiologic research, and its discontents (Bennette & Vickers; 2012)
16. **Dichotomizing continuous variables** in statistical analysis: **a practice to avoid** (Dawson & Weiss; 2012)
17. The **danger of dichotomizing continuous variables**: A visualization (Kuss 2013)
18. The "**anathema**" of **arbitrary categorization of continuous predictors** (Vintzileos et al; 2014)
19. Ophthalmic statistics note: **the perils of dichotomising continuous variables** (Cumberland et al 2014)

Source: Gary Collins, <https://bit.ly/30Kr3W2>

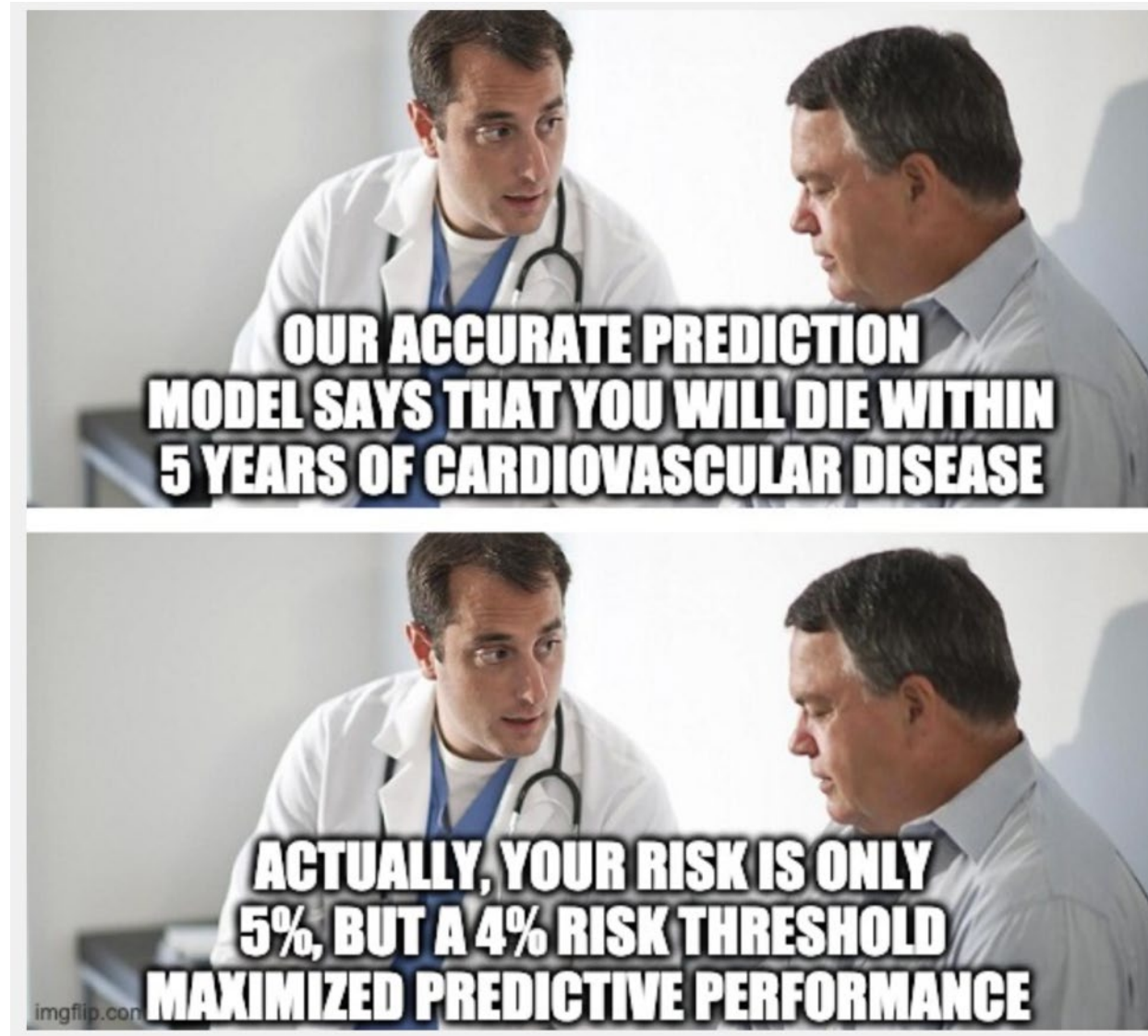
Unnecessary dichotomizations of predictors

- **Biological implausible** step-functions in predicted risk
- **Source of overfitting** when cut-off is chosen based on maximizing predictive performance
- **Loss of information** (e.g. Ensor et al. show dichotomizing BMI was equivalent to throwing away 1/3 of the data)

Dichotomization/categorization remains **very prevalent**

- Wynants et al. 2020 (COVID prediction models): **48%**
- Collins et al. 2011 (Diabetes T2 prediction models): **63%**
- Mallett et al. 2010 (Prognostic models in cancer): **70%**

Unnecessary dichotomizations of predictionsu

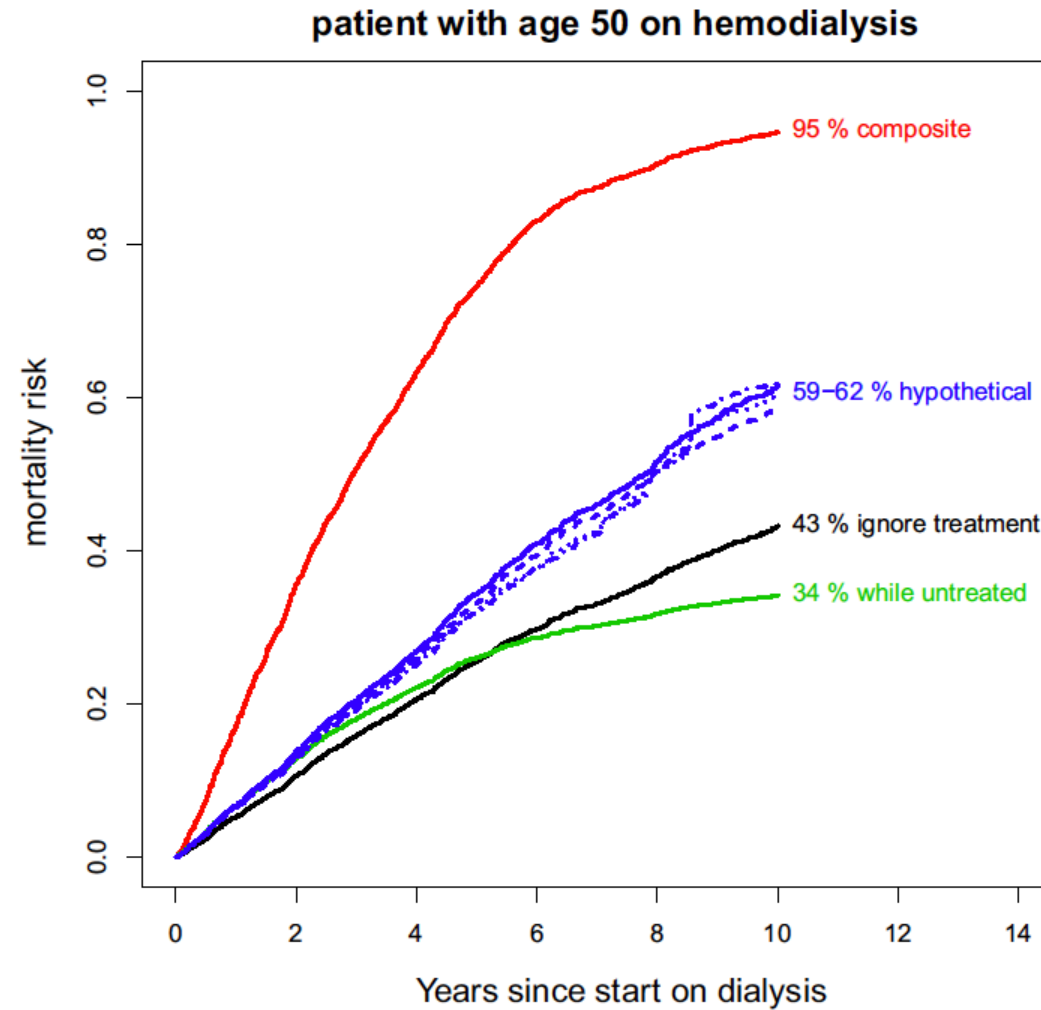


Predictimands

Table 1 Overview of the four strategies of dealing with treatment initiation after baseline

Strategy	Estimand	Example
Ignore treatment	Risk of the event, regardless of treatment	Risk of cardiovascular events where some patients will initiate statins according to routine-care prescriptions
Composite	Risk of the event or treatment initiation	Risk of a composite of cardiovascular death, myocardial infarction and treatment with revascularisation (PCI or CABG)
While untreated	Risk of the event occurring before treatment is started	Risk of dying while on the waiting list for a liver transplant
Hypothetical	Risk of the event if treatment were never started	Risk of a natural pregnancy without IVF treatment

Predictimands



Van Geloven et al. EJE, 2020

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 - Applied example: ADNEX model
 - Future perspective: machine learning and AI

Overfitting =

**PUBLIC
ENEMY
NUMBER
ONE....**



**THE BEST WAY TO
EXPLAIN OVERFITTING**

Classical consequence of overfitting

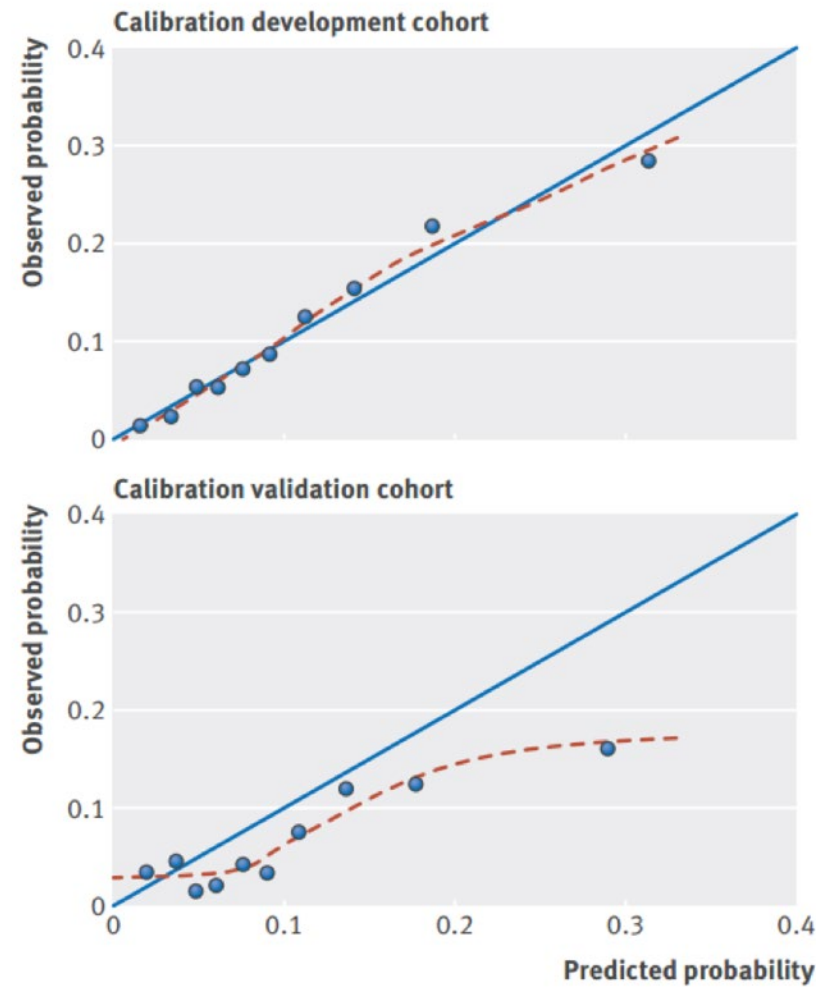


Fig 1 | Calibration in development and validation cohorts

Bell et al. BMJ, 2015

Shortest routes to overfitting



Journal of Clinical Epidemiology ■ (2017) ■

Journal of
Clinical
Epidemiology

ORIGINAL ARTICLE

Poor performance of clinical prediction models: the harm of commonly applied methods

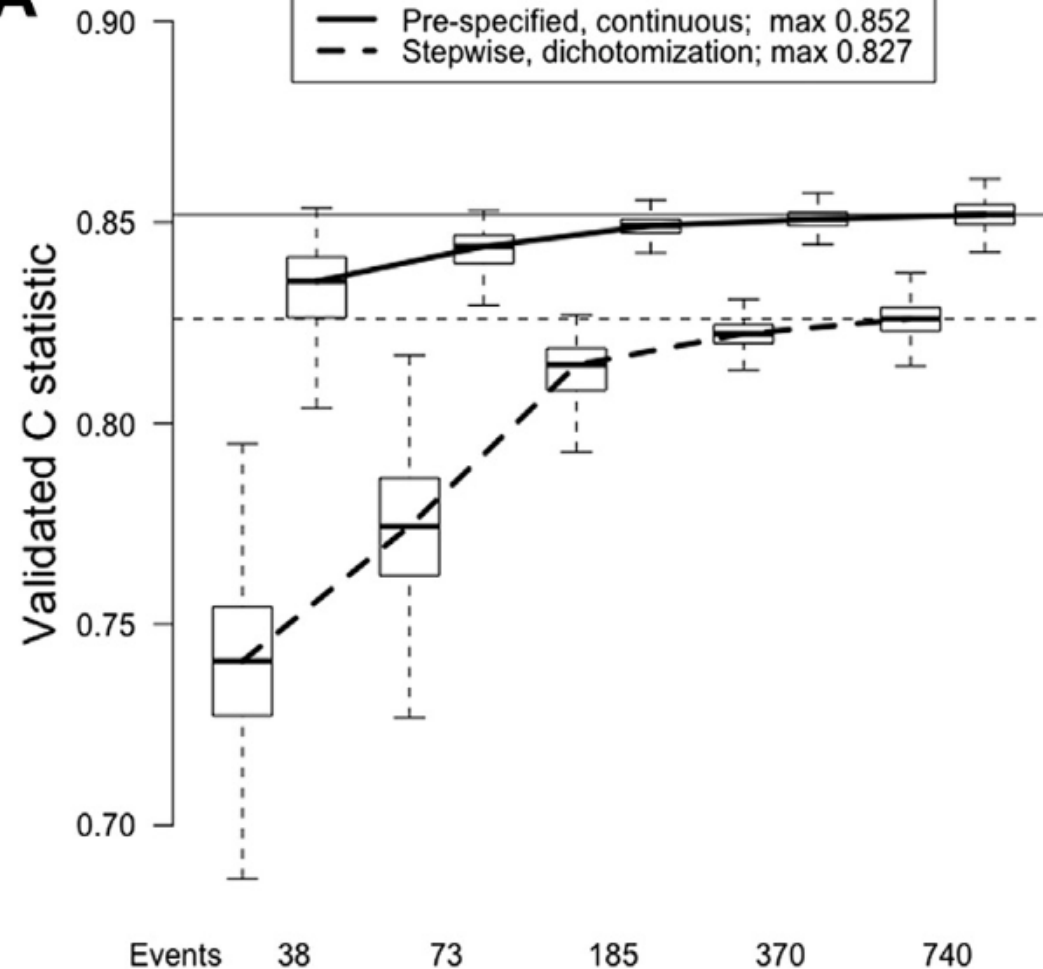
Ewout W. Steyerberg^{a,b,*}, Hajime Uno^c, John P.A. Ioannidis^{d,e,f,g}, Ben van Calster^{a,h},
Collaborators

Study Design and Setting: A previously published model predicted the likelihood of having a mutation in germline DNA mismatch repair genes at the time of diagnosis of colorectal cancer. This model was based on a cohort where 38 mutations were found among 870 participants, with validation in an independent cohort with 35 mutations. The modeling strategy included stepwise selection of predictors from a pool of over 37 candidate predictors and dichotomization of continuous predictors. We simulated this strategy in small subsets of a large contemporary cohort (2,051 mutations among 19,866 participants) and made comparisons to other modeling approaches. All models were evaluated according to bias and discriminative ability (concordance index, *c*) in independent data.

Shortest routes to overfitting



A



Journal of
Clinical
Genetics

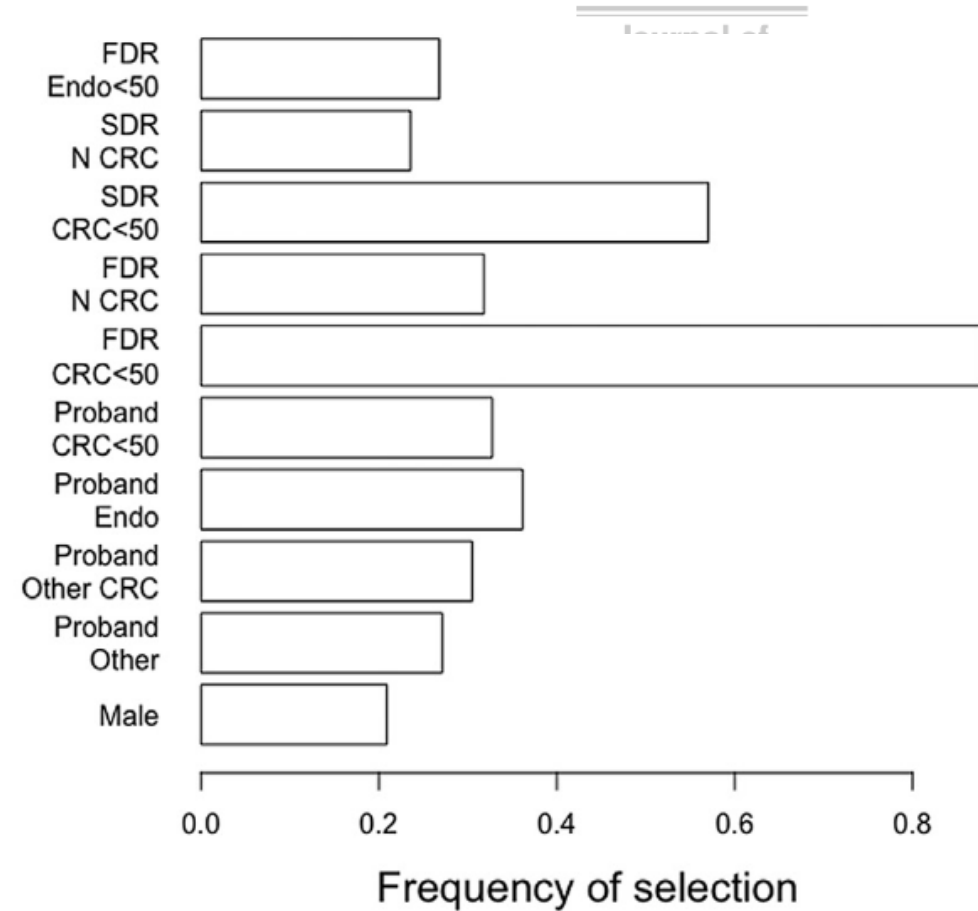
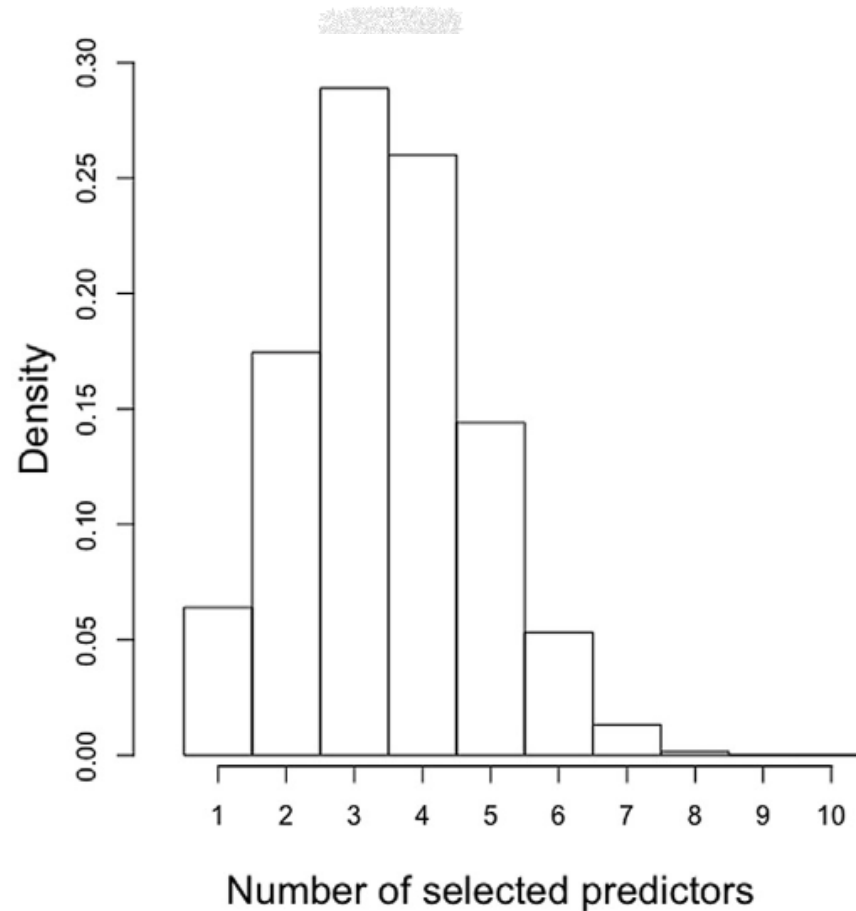
n

ster^{a,h},

Study Design and Setting: A study of DNA mismatch repair genes at the time of diagnosis in participants, with validation in a separate cohort from a pool of over 37 candidate genes. The study was conducted in a large contemporary cohort (2,051 participants) and the models were evaluated according to bias.

on in germline DNA mismatch repair gene mutations were found among 870 participants. Stepwise selection of predictors was used in small subsets of a modeling approaches. All models

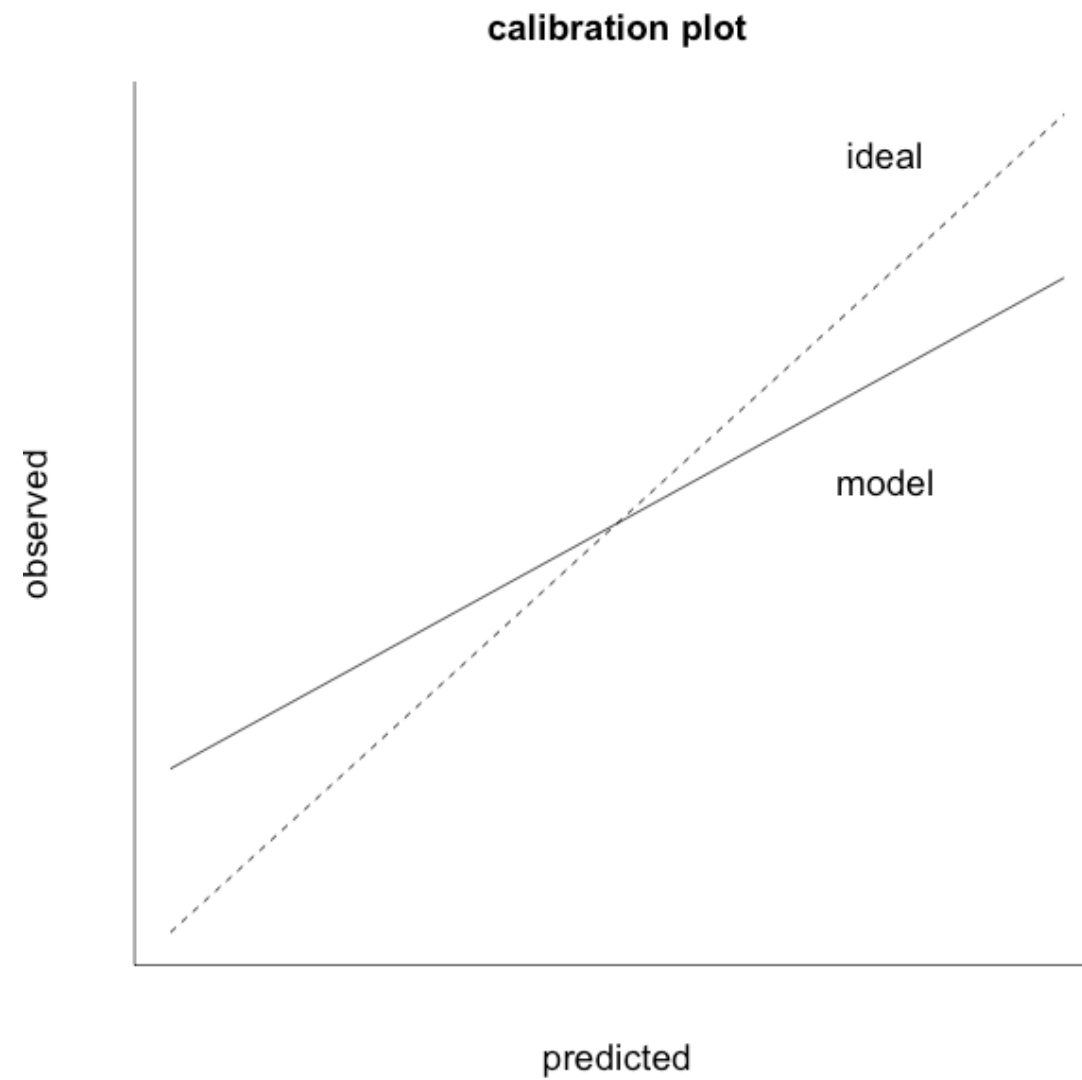
Shortest routes to overfitting

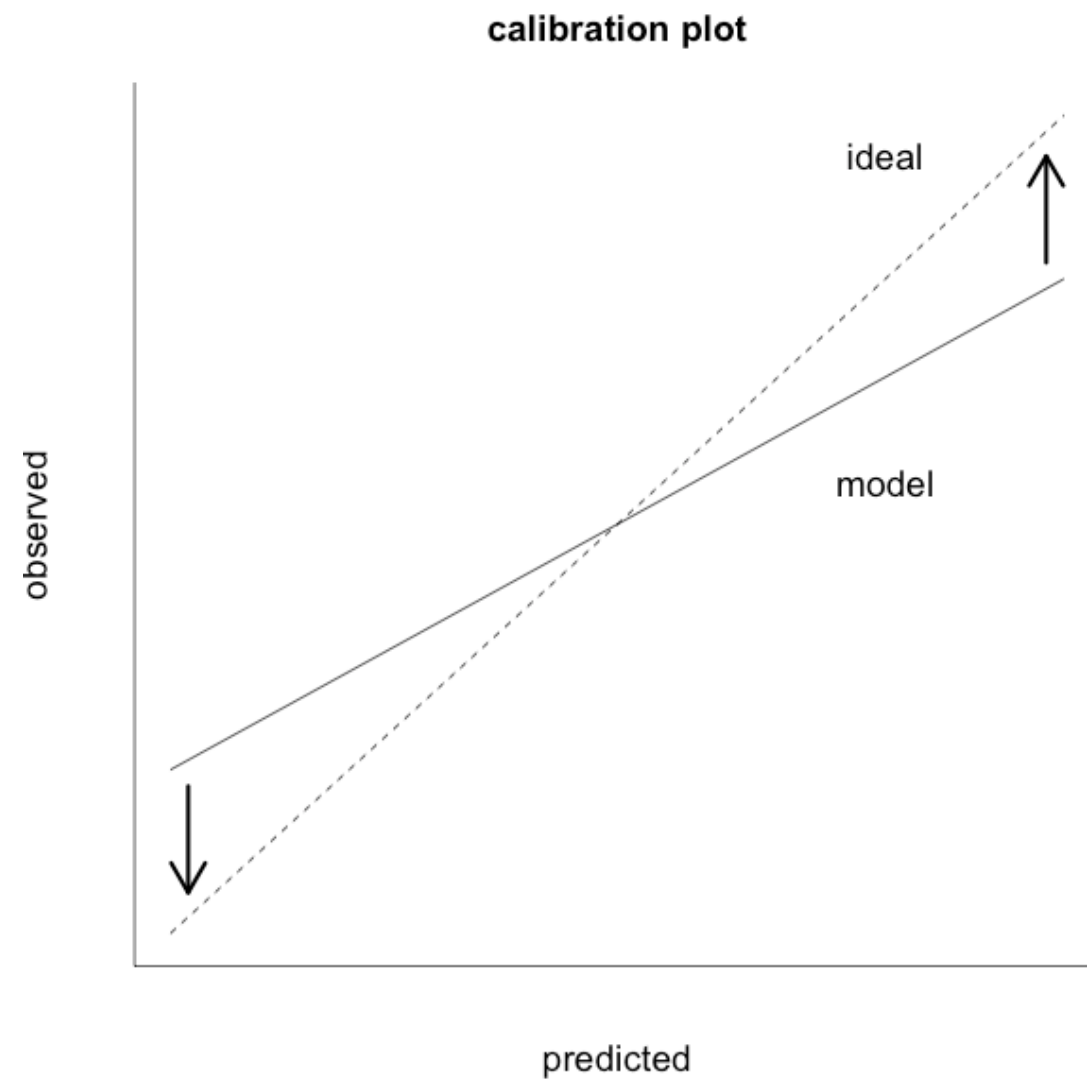


Routes to preventing overfitting

- **Sample size**: gather data on a sufficient* number of individuals
 - Use it all on model development (avoid training-test splitting)
- Careful candidate **predictors preselection**
 - Especially if data are small
- **Avoid** and be aware of the **winner's curse**
 - Data driven variable selection, model selection
- **Avoid unnecessary dichotomizations**
- Regression **shrinkage** / penalization / regularization

* Topic of 4th topic of this talk





Shrinkage

Post-estimation shrinkage factor estimation

- Van Houwelingen & Le Cessie, 1990: uniform shrinkage factor, S_{VH}
- Sauerbrei 1999: parameterwise shrinkage factors

Regularized regression (shrinkage during estimation)

- Ridge regression: L2-penalty on regression coefficient
- Lasso: L1 penalty
- Elastic net: L1 and L2 penalty

Shrinkage

Post-estimation shrinkage factor estimation

$$\Pr(Y = 1) = \text{expit}[\beta_0^* + S_{VH}(\beta_1 X_1 + \dots + \beta_P X_P)]$$

Regularized regression (shrinkage during estimation)

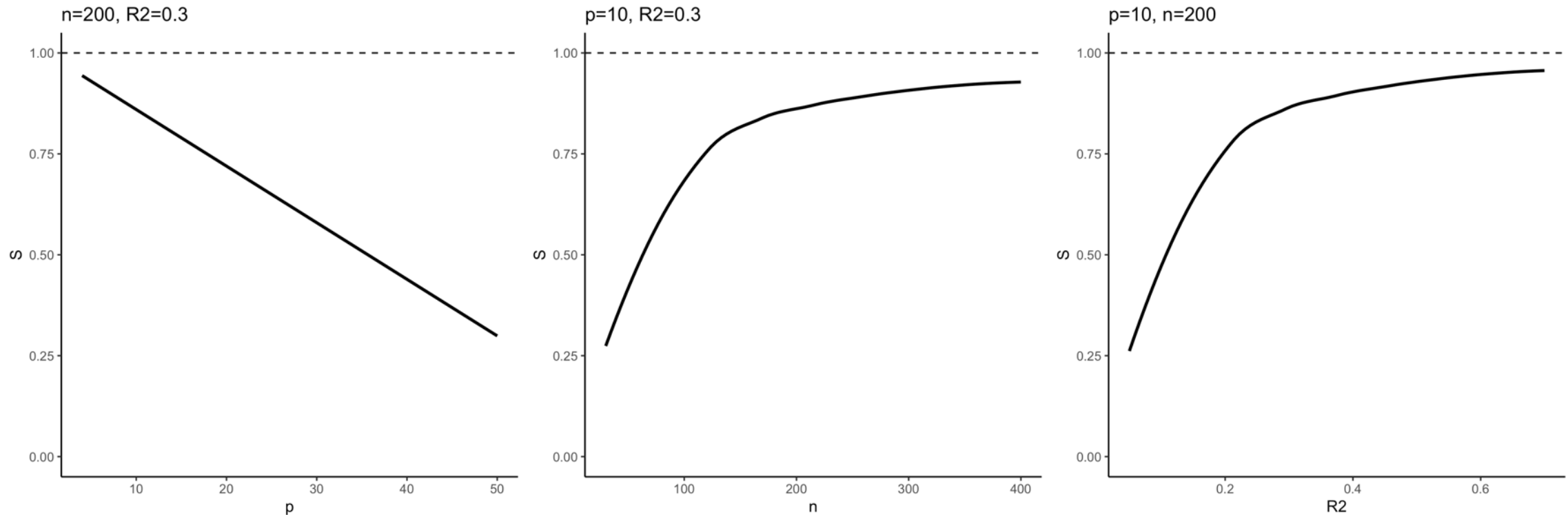
$$\text{Ln}\mathcal{L}_p = \text{Ln}\mathcal{L}_{ml} - \lambda \left[(1 - \alpha) \sum_{p=1}^P \beta_p^2 + \alpha \sum_{p=1}^P |\beta_p| \right]$$

Ridge regression: $\alpha = 0$, Lasso: $\alpha = 1$, Elastic net $0 < \alpha < 1$

Uniform shrinkage factor

$$S_{VH} = 1 + \frac{p}{n \ln(1 - R^2_{CS_app})}$$

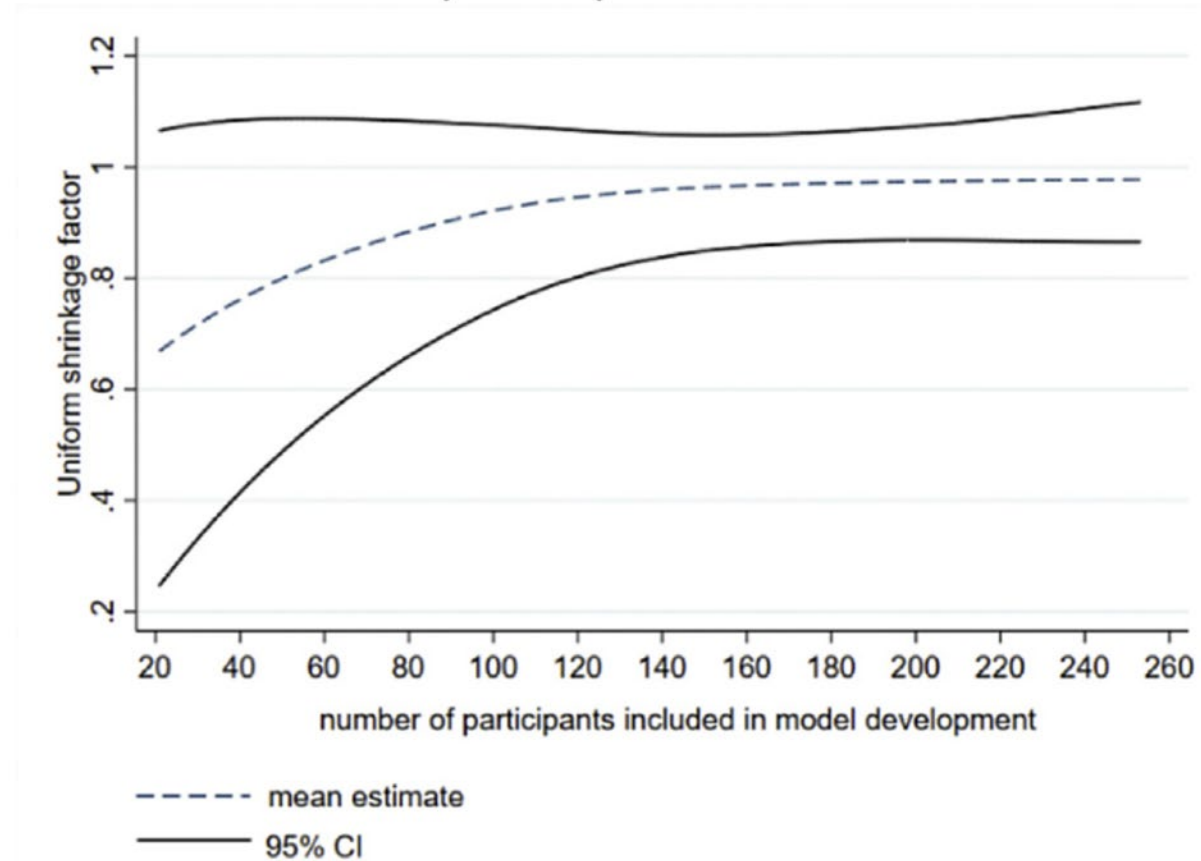
Shrinkage < 1 means overfitting
lower values = more overfitting



Formula from Riley et al.

Shrinkage/tuning/penalization hard to estimate

B R_{app}^2 is 0.56 for model B when using the maximum of 253 participants



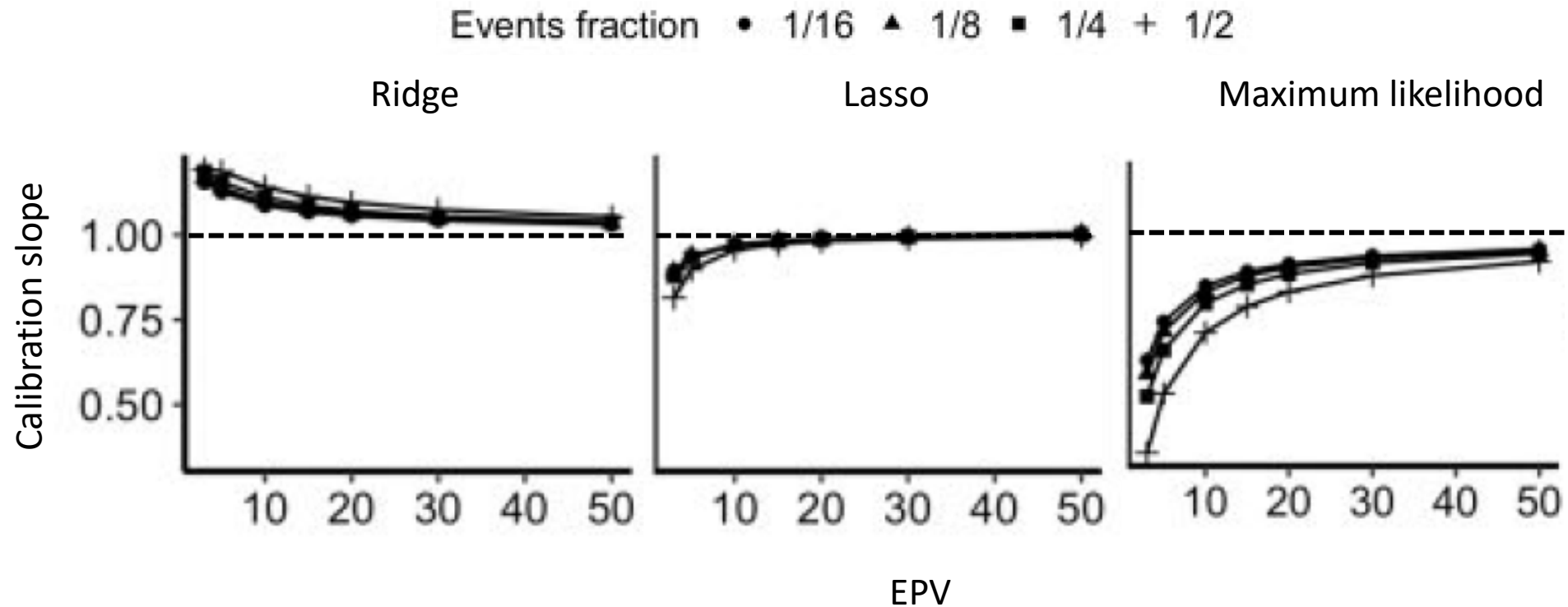
Statistica Neerlandica (2001) Vol. 55, nr. 1, pp. 17–34

Shrinkage and penalized likelihood as methods to improve predictive accuracy

J. C. van Houwelingen*

“shrinkage works on the average but may fail in the particular unique problem on which the statistician is working.”

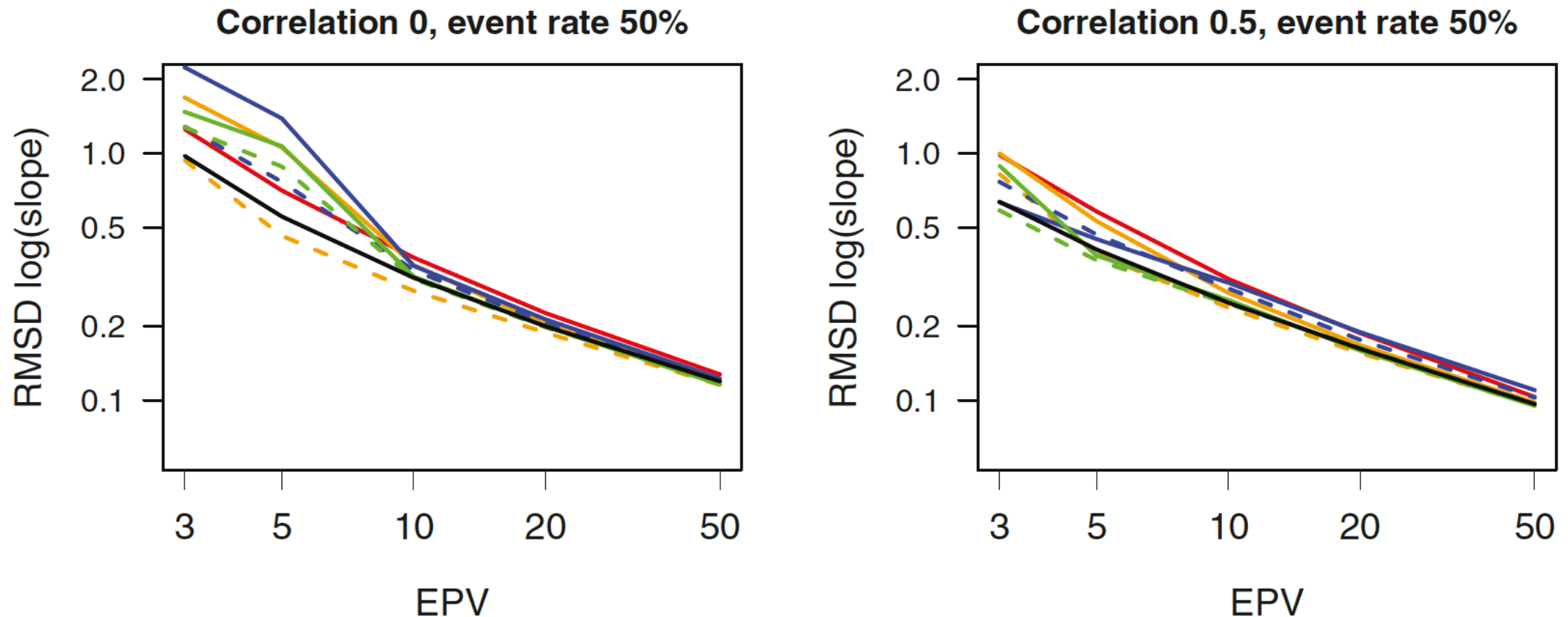
Shrinkage works on the average



Simulation results, averaged over 4k scenarios and 20M datasets

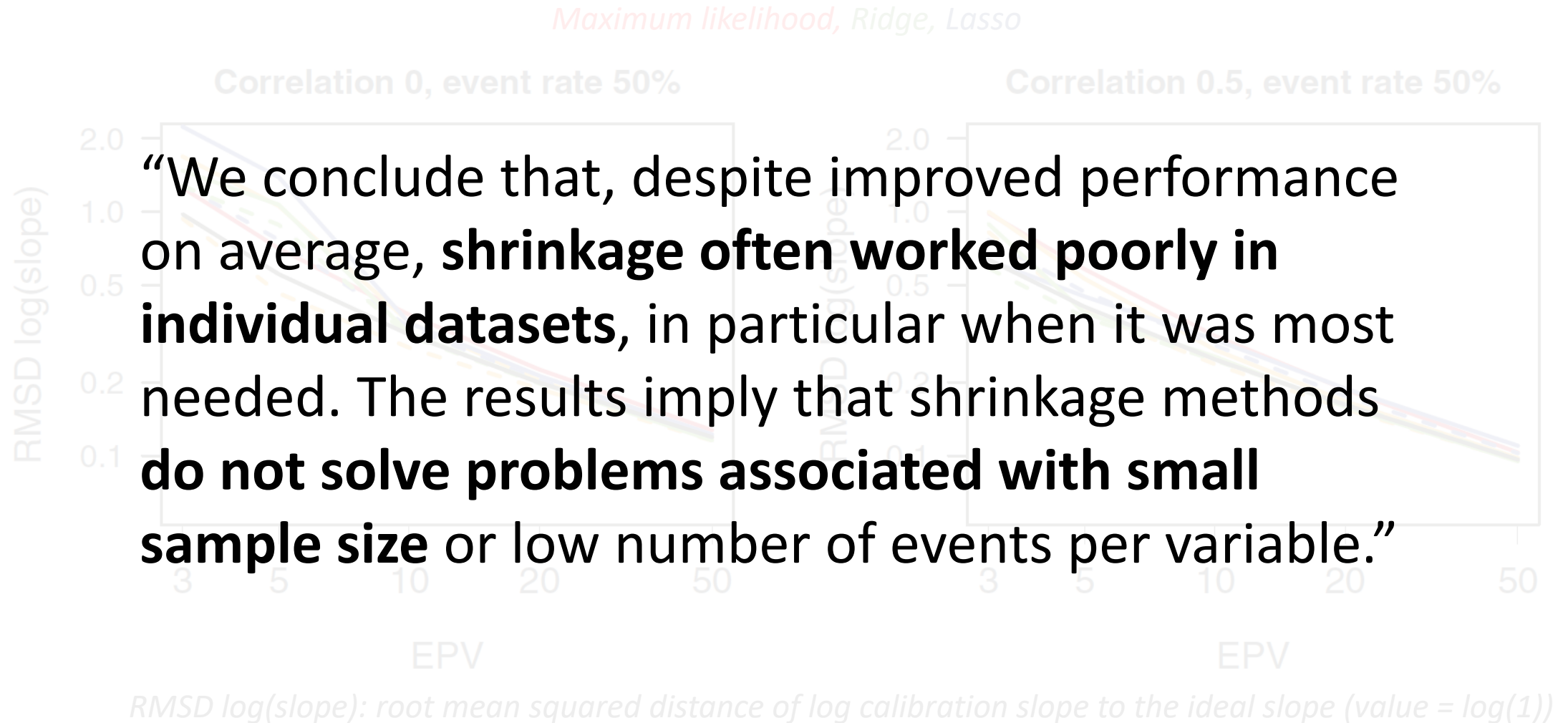
Shrinkage may not work on a particular dataset

Maximum likelihood, Ridge, Lasso

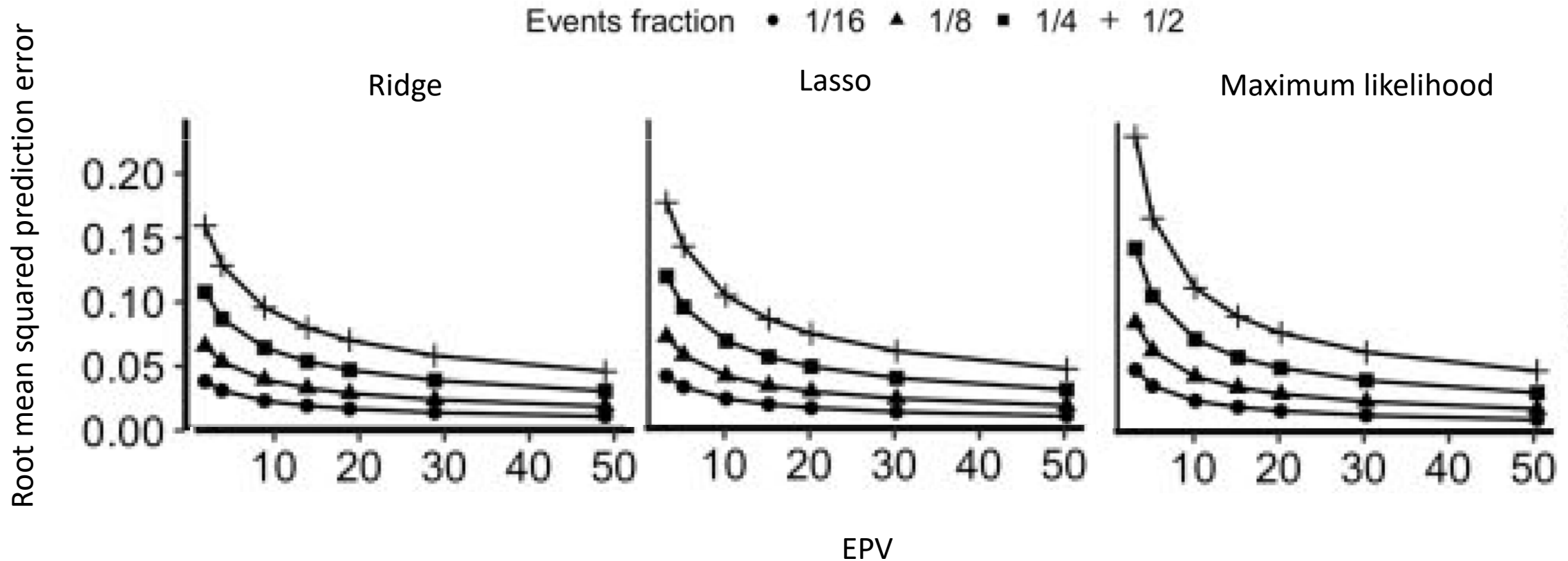


RMSD log(slope): root mean squared distance of log calibration slope to the ideal slope (value = log(1))

Shrinkage may not work on a particular dataset



What about the variance in estimated risk?



Sources of prediction error

$$Y = f(x) + \varepsilon$$

For a model k the **expected test prediction error** is:

$$\sigma^2 + \underbrace{\text{bias}^2(\hat{f}_k(x)) + \text{var}(\hat{f}_k(x))}_{\text{Mean squared prediction error}}$$

Irreducible error

$\approx$

What we don't model

$\approx$

How we model

(with $E(\varepsilon) = 0$, $\text{var}(\varepsilon) = \sigma^2$, values in x are not random)

See equation 2.46 in Hastie et al., the elements of statistical learning, <https://stanford.io/2voWjra>

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You get what you pay for



Sample size considerations

- Sample **size matters** when designing a **study of any kind**
- When designing a prediction study precedes data collection (*prospective*)
 - **How many** units/individuals/events do I need to collect data on?
- When designing a prediction study on existing data (*retrospective*)
 - Is my dataset **large enough** to build a model?
 - How much **model complexity** (e.g. number predictors considered) can I afford?



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One in ten rule

From Wikipedia, the free encyclopedia

In [statistics](#), the **one in ten rule** is a [rule of thumb](#) for how many predictors can be derived from data when doing [regression analysis](#) (in particular [proportional hazards models](#) and [logistic regression](#)) without risk of [overfitting](#). The rule states that one predictive variable can be studied for every ten events.^{[1][2][3][4]} It is not applicable to ordinary least squares [linear regression](#), where it is suggested that as few as two events per predictor are sufficient.^[5]

The origin of the 1 in 10 rule

J Clin Epidemiol Vol. 49, No. 12, pp. 1373–1379, 1996
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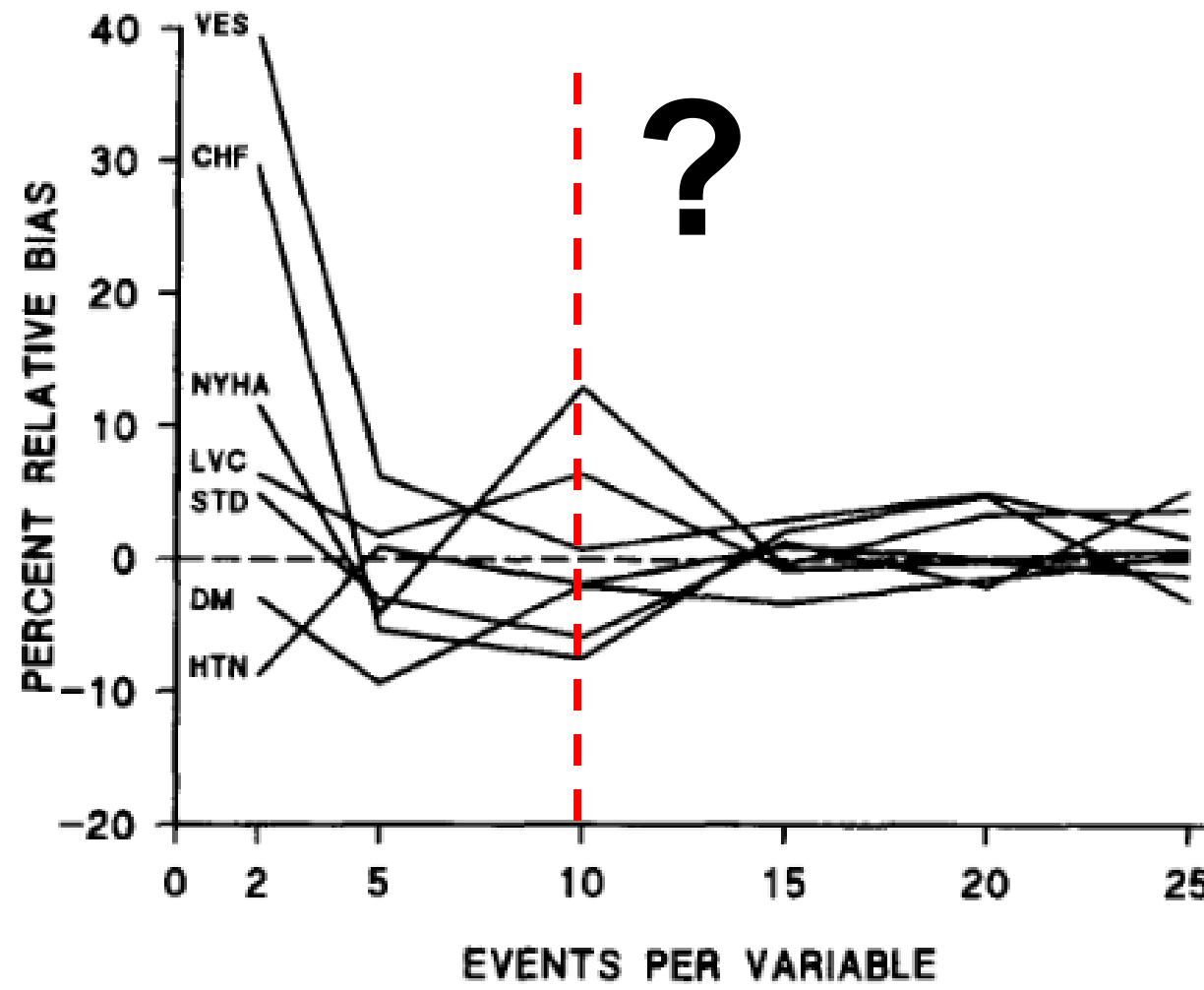


S0895-4356/96/\$15.00
PII S0895-4356(96)00236-3

A Simulation Study of the Number of Events per Variable in Logistic Regression Analysis

Peter Peduzzi,^{1,4,} John Concato,^{2,3} Elizabeth Kemper,^{1,4} Theodore R. Holford,⁴ and
Alvan R. Feinstein^{2,3,4}*

“For EPV values of 10 or greater, no major problems occurred. For **EPV values less than 10, however, the regression coefficients were **biased in both positive and negative directions**”**



Peduzzi et al. JCE, 1996

More simulation studies

J Clin Epidemiol Vol. 49, No. 12, pp. 1373–1379, 1996
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S0895-4356(96)\$15.00
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For EPV values of 10 or greater, no major problems
citations: 5,736



Journal of Clinical Epidemiology 64 (2011) 993–1000

Performance of logistic regression modeling: beyond the number of events per variable, the role of data structure

Delphine S. Courvoisier^{a,b,*}, Christophe Combescure^{a,b}, Thomas Agoritsas^{a,b},
Angèle Gayet-Ageron^{a,b}, Thomas V. Perneger^{a,b}

“substantial problems even if the number of EPV exceeds 10”
citations: 216

Citations based on Google Scholar, Oct 30 2020



American Journal of Epidemiology
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DOI: 10.1093/aje/kwk052

Original Contribution

Relaxing the Rule of Ten Events per Variable in Logistic and Cox Regression

Eric Vittinghoff and Charles E. McCulloch

“a minimum of 10 EPV [...] may be too conservative”
citations: 2,438

**Journal of
Clinical
Epidemiology**

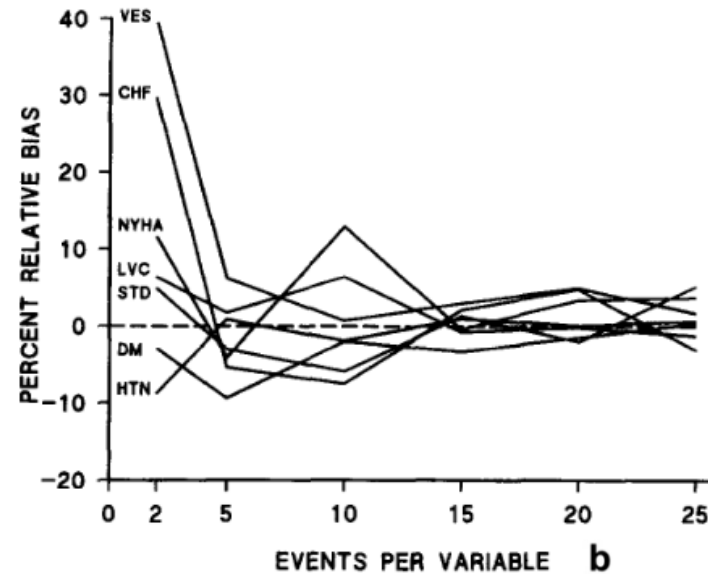
More simulation studies

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Peter

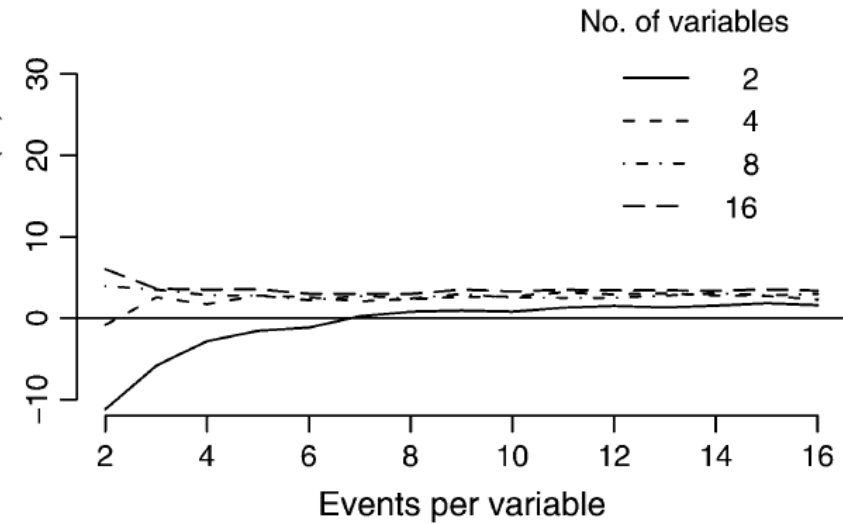
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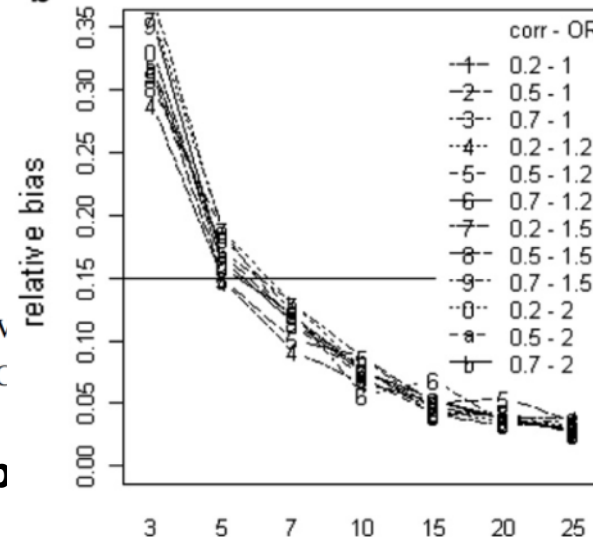
EVENTS PER VARIABLE **b**



ELSEVIER

Performance
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Journal of
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Epidemiology

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Agoritsas^{a,b},

“PV exceeds 10”

Citations based on Google Scholar, Oct 30 2020

Log(odds) is consistent but finite sample biased

Biometrika (1967), 54, 1 and 2, p. 181

Printed in Great Britain

181

TECHNOMETRICS ©, VOL. 21, NO. 1, FEBRUARY 1979

On the bias of various estimators of the logit and its variance with application to quantal bioassay

BY JOHN J. GART AND JAMES R. ZWEIFEL

National Cancer Institute, Bethesda, Md

BIOMETRICS 40, 421–435
June 1984

Small-Sample Bias of Point Estimators of the Odds Ratio from Matched Sets

Nicholas P. Jewell

Department of Biomedical and Environmental Health Sciences, School of Public Health,
University of California, Berkeley, California 94720, U.S.A.

STATISTICS IN MEDICINE
Statist. Med. 2006; 25:4216–4226
Published online 6 September 2006 in Wiley InterScience
(www.interscience.wiley.com) DOI: 10.1002/sim.2687



A comparative investigation of methods for logistic regression with separated or nearly separated data

Georg Heinze*,†

Logistic Discrimination and Bias Correction in Maximum Likelihood Estimation

J. A. Anderson

and

S. C. Richardson

J. R. Statist. Soc. B (1991)
53, No. 3, pp. 629–643

Bias Correction in Generalized Linear Models

By GAUSS M. CORDEIRO†

and

PETER McCULLAGH

Universidade Federal de Pernambuco, Brazil

University of Chicago, USA

BMC Medical Research Methodology



Research article

Open Access

Bias in odds ratios by logistic regression modelling and sample size

Szilard Nemes*¹, Junmei Miao Jonasson¹, Anna Genell¹ and
Gunnar Steineck^{1,2}

STATISTICS IN MEDICINE, VOL. 2, 71–78 (1983)

BIAS CORRECTION IN MAXIMUM LIKELIHOOD LOGISTIC REGRESSION

ROBERT L. SCHAEFER

Department of Mathematics and Statistics, Miami University, Oxford, Ohio, U.S.A.

$$\begin{aligned} \text{bias}(\hat{\beta}) \approx & (-1/2) (\mathbf{X}^T \mathbf{V} \mathbf{X})^{-1} \mathbf{X}^T \{ \text{trace}(\mathbf{X} (\mathbf{X}^T \mathbf{V} \mathbf{X})^{-1} E^k (\mathbf{X}^T \mathbf{V} \mathbf{X})^{-1} \mathbf{X}^T \text{Var}(\varepsilon)) \} \\ & (-1/2) (\mathbf{X}^T \mathbf{V} \mathbf{X})^{-1} \mathbf{X}^T \mathbf{V} \{ (1 - 2\pi_k) \mathbf{x}_k^T (\mathbf{X}^T \mathbf{V} \mathbf{X})^{-1} \mathbf{x}_k \} \end{aligned} \quad (6)$$

RESEARCH ARTICLE

Open Access



No rationale for 1 variable per 10 events criterion for binary logistic regression analysis

Maarten van Smeden^{1*} , Joris A. H. de Groot¹, Karel G. M. Moons¹, Gary S. Collins², Douglas G. Altman², Marinus J. C. Eijkemans¹ and Johannes B. Reitsma¹

- Examine the **reasons** for substantial **differences** between the earlier EPV simulation studies
- Evaluate a possible **solution** to reduce the **finite sample bias**

RESEARCH ARTICLE

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RESEARCH ARTICLE

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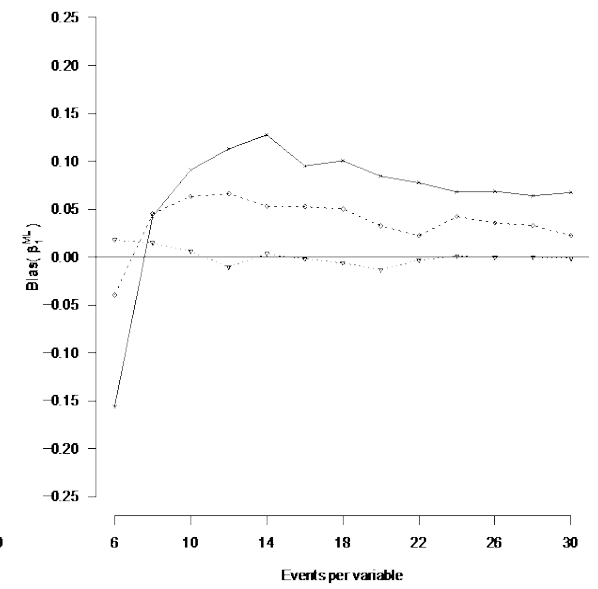
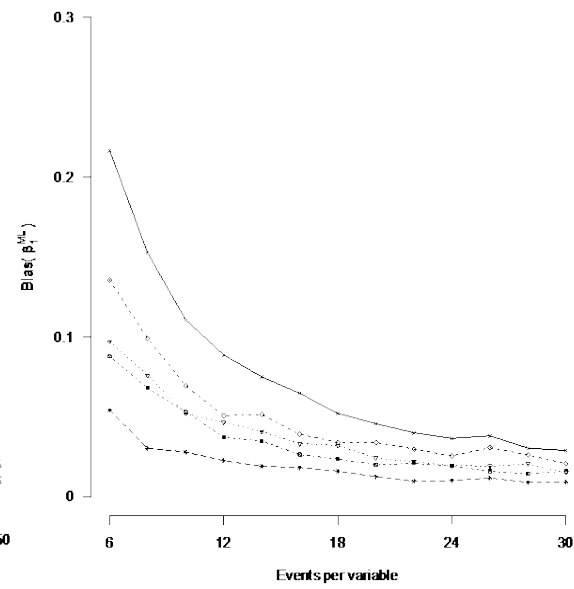
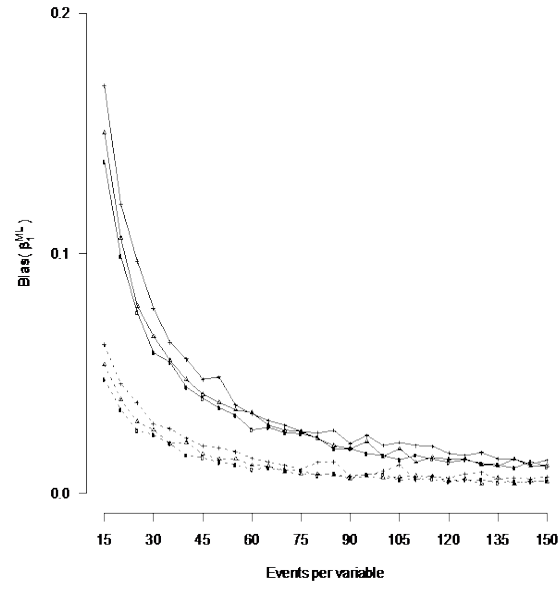
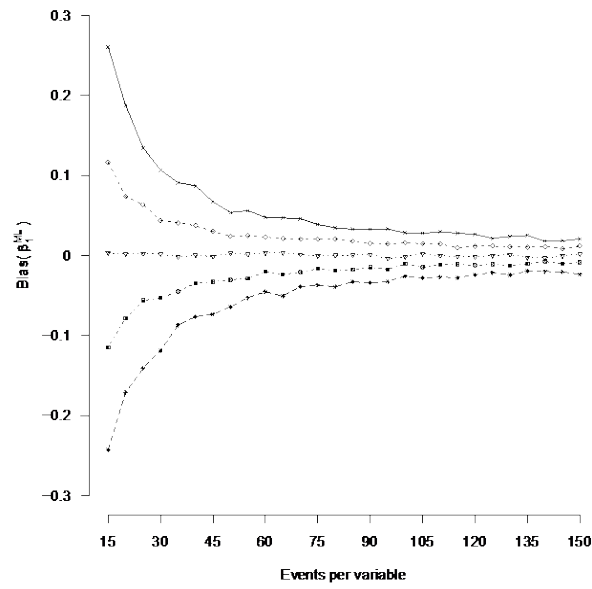
Bias reduction of maximum likelihood estimates

BY DAVID FIRTH

Department of Mathematics, University of Southampton, SO9 5NH, U.K.

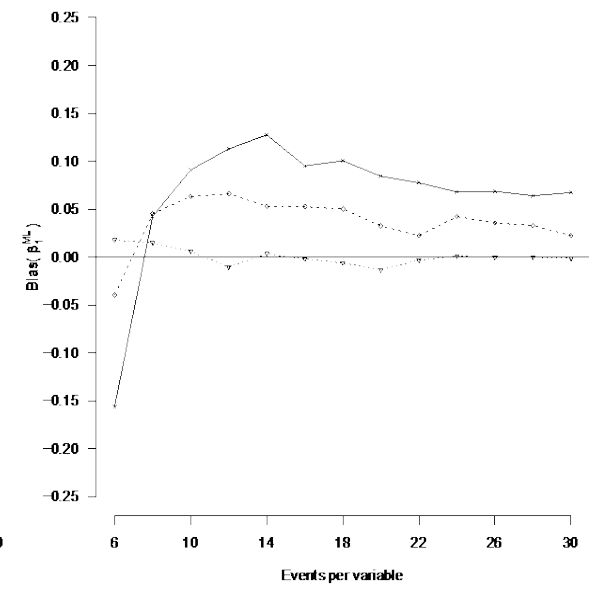
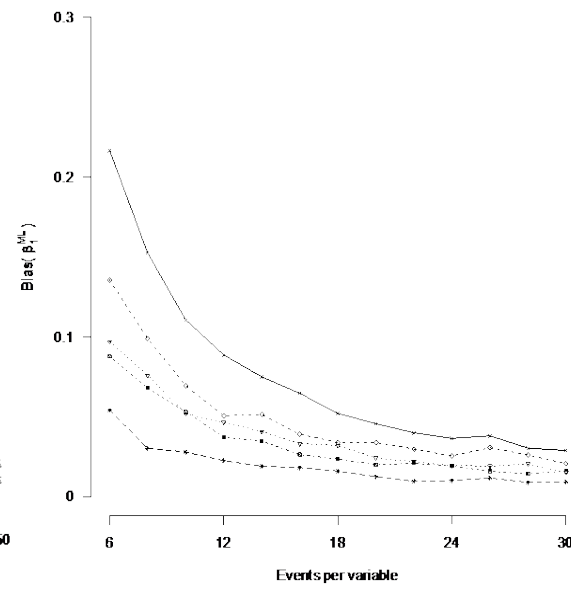
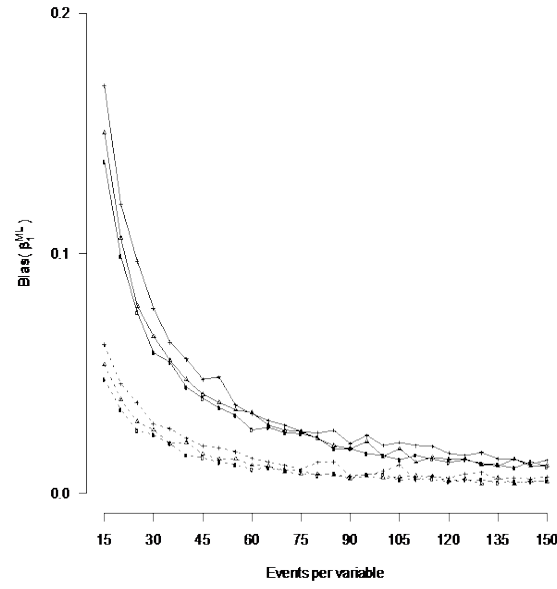
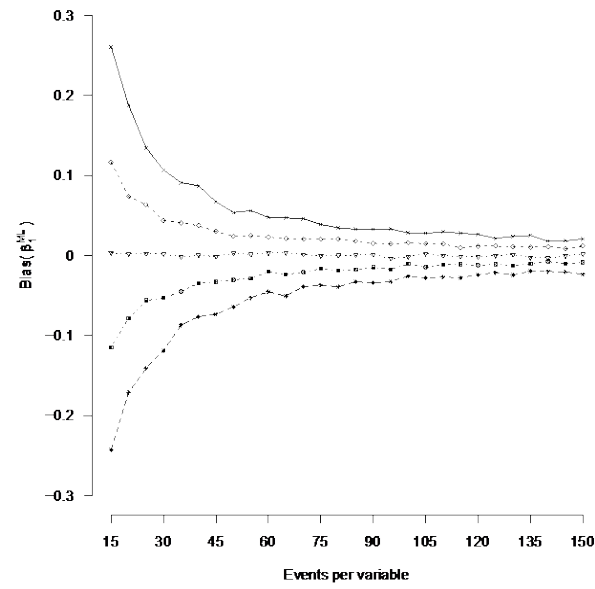
- Firth's "correction" aims to **reduce finite sample bias** in maximum likelihood estimates, applicable to logistic regression
- It makes clever use of the "Jeffries prior" (from Bayesian literature) to penalize the log-likelihood, which **shrinks** the estimated coefficients
- Nice theoretical justifications

Standard

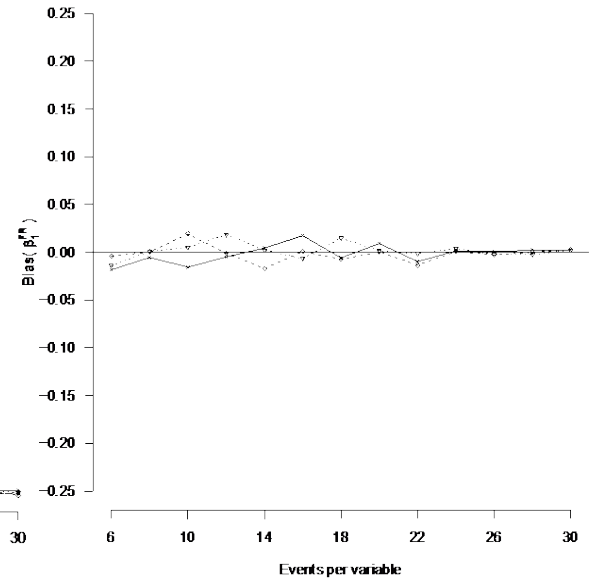
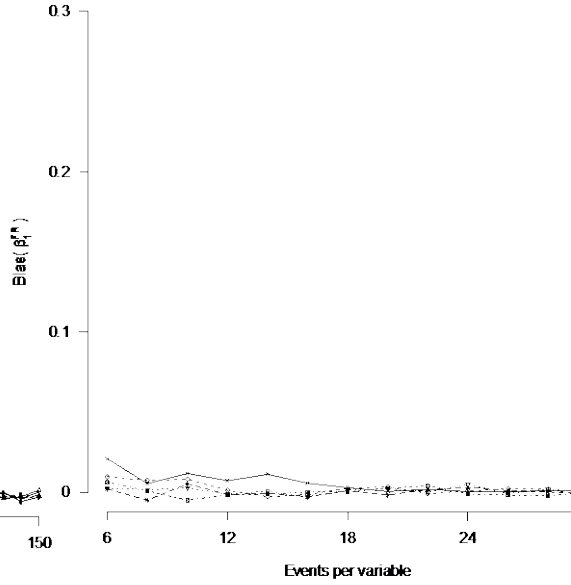
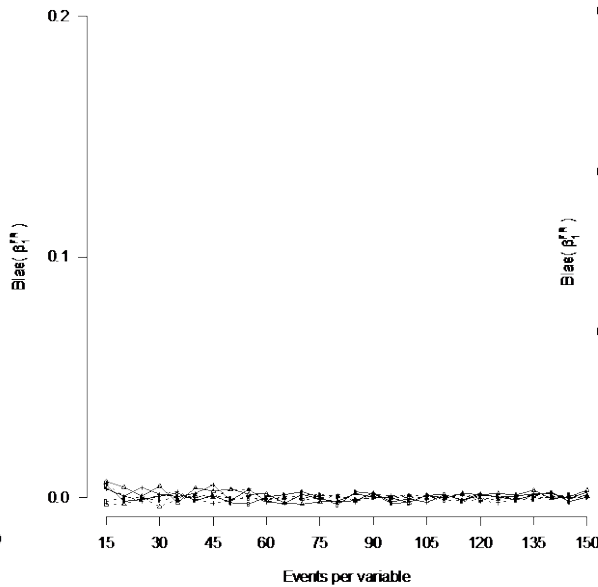
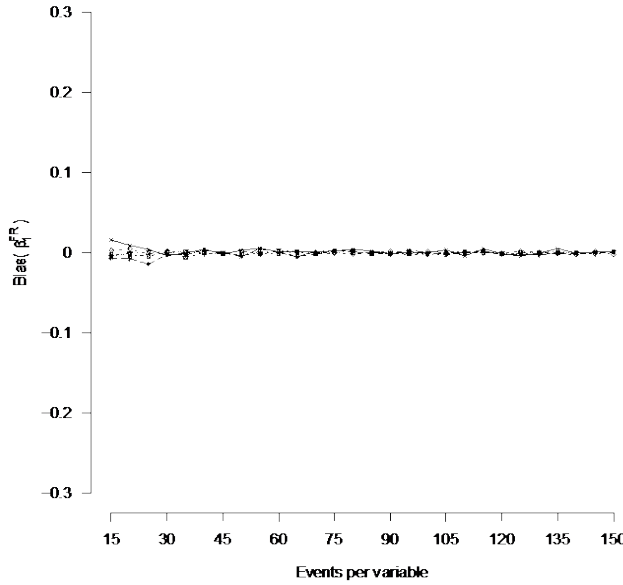


Averaged over 465 simulation conditions with 10,000 replications each

Standard



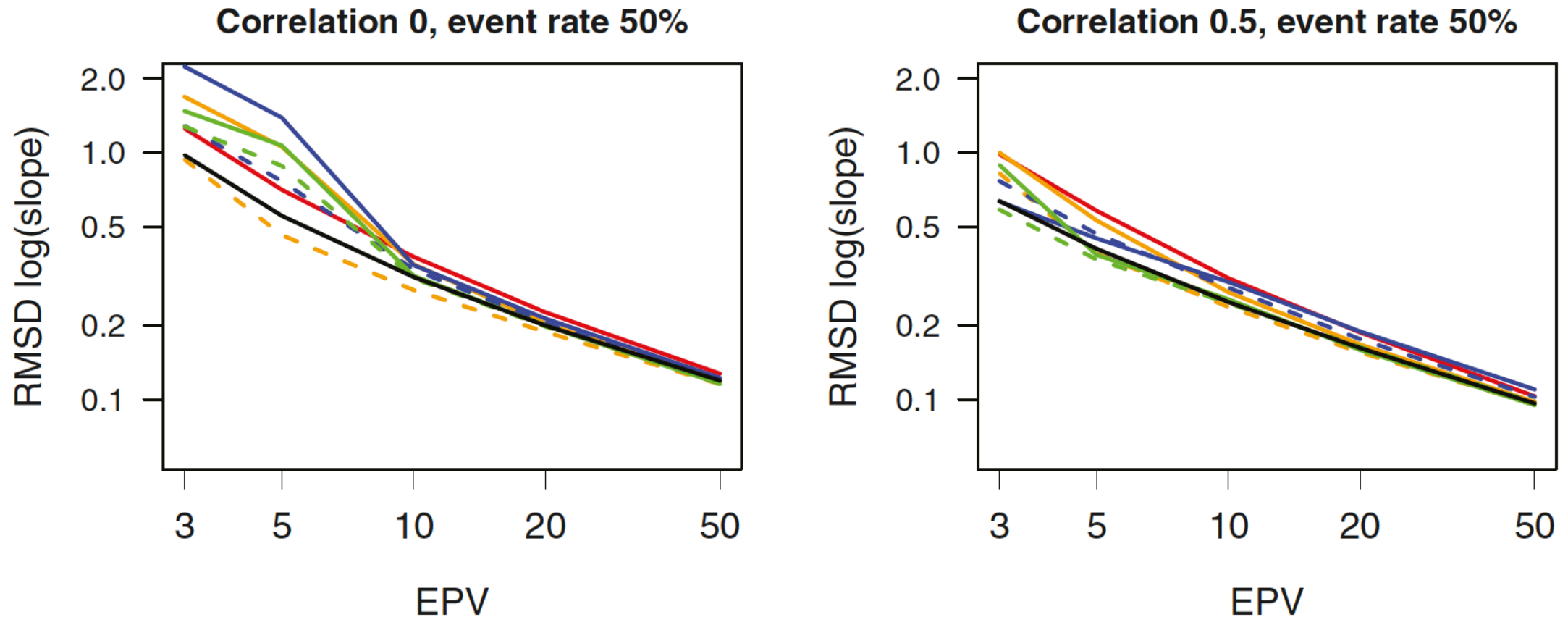
Firth's correction



Averaged over 465 simulation conditions with 10,000 replications each

Firth's correction and predictive performance

Maximum likelihood, Firth's correction



RMSD log(slope): root mean squared distance of log calibration slope to the ideal slope (value = log(1))

Beyond events per variable

- Sample size has a **big influence** on the performance of prediction models
 - Giving appropriate weights to predictors is not a simple task
- Challenge is **to avoid overfitting and lack of precision** in the predictions
 - Requires adequate sample size
- How much data is sufficient? Depends
 - **Model complexity** (e.g. # predictors, EPV)
 - **Signal:noise ratio** (e.g. R^2 or C-index)
 - **Performance required** (e.g. high vs low stake medical decisions)



RESEARCH ARTICLE

WILEY Statistics
in Medicine

Sample size for binary logistic prediction models: Beyond events per variable criteria

Maarten van Smeden,¹  Karel GM Moons,¹ Joris AH de Groot,¹ Gary S Collins,²  Douglas G Altman,² Marinus JC Eijkemans¹ and Johannes B Reitsma¹

Received: 4 February 2018 | Revised: 13 September 2018 | Accepted: 13 September 2018
DOI: 10.1002/sim.7993

RESEARCH ARTICLE

Minimum sample size for developing a multivariable prediction model: Part I – Continuous outcomes

Richard D. Riley¹  | Kym I.E. Snell¹ | Joie Ensor¹  | Danielle L. Burke¹  | Frank E. Harrell Jr² | Karel G.M. Moons³ | Gary S. Collins⁴ 

WILEY Statistics
in Medicine

Received: 4 February 2018 | Revised: 13 September 2018 | Accepted: 13 September 2018
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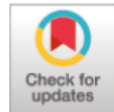
RESEARCH ARTICLE

WILEY Statistics
in Medicine

Minimum sample size for developing a multivariable prediction model: PART II - binary and time-to-event outcomes

Richard D Riley¹  | Kym IE Snell¹ | Joie Ensor¹  | Danielle L Burke¹  | Frank E Harrell Jr² | Karel GM Moons³ | Gary S Collins⁴ 

updates



RESEARCH METHODS & REPORTING

Calculating the sample size required for developing a clinical prediction model

Clinical prediction models aim to predict outcomes in individuals, to inform diagnosis or prognosis in healthcare. Hundreds of prediction models are published in the medical literature each year, yet many are developed using a dataset that is too small for the total number of participants or outcome events. This leads to inaccurate predictions and consequently incorrect healthcare decisions for some individuals. In this article, the authors provide guidance on how to calculate the sample size required to develop a clinical prediction model.

Richard D Riley *professor of biostatistics*¹, Joie Ensor *lecturer in biostatistics*¹, Kym I E Snell *lecturer in biostatistics*¹, Frank E Harrell Jr *professor of biostatistics*², Glen P Martin *lecturer in health data sciences*³, Johannes B Reitsma *associate professor*⁴, Karel G M Moons *professor of clinical epidemiology*⁴, Gary Collins *professor of medical statistics*⁵, Maarten van Smeden *assistant professor*^{4 5 6}

Our proposal

- Calculate sample size that is needed to
 - minimise potential **overfitting**
 - estimate **probability** (risk) precisely
- Sample size formula's for
 - Continuous outcomes
 - Time-to-event outcomes
 - **Binary outcomes** (focus today)

Example

Development and validation of the ISARIC 4C Deterioration model for adults hospitalised with COVID-19: a prospective cohort study

Rishi K Gupta, Ewen M Harrison, Antonia Ho, Annemarie B Docherty, Stephen R Knight, Maarten van Smeden, Ibrahim Abubakar, Marc Lipman, Matteo Quartagno, Riinu Pius, Iain Buchan, Gail Carson, Thomas M Drake, Jake Dunning, Cameron J Fairfield, Carrol Gamble, Christopher A Green, Sophie Halpin, Hayley E Hardwick, Karl A Holden, Peter W Horby, Clare Jackson, Kenneth A Mclean, Laura Merson, Jonathan S Nguyen-Van-Tam, Lisa Norman, Piero L Olliaro, Mark G Pritchard, Clark D Russell, James Scott-Brown, Catherine A Shaw, Aziz Sheikh, Tom Solomon, Cathie Sudlow, Olivia V Swann, Lance Turtle, Peter J M Openshaw*, J Kenneth Baillie*, Malcolm G Semple*, Mahdad Noursadeghi*, on behalf of the ISARIC4C Investigators

Summary

Background Prognostic models to predict the risk of clinical deterioration in acute COVID-19 cases are urgently required to inform clinical management decisions.

Methods We developed and validated a multivariable logistic regression model for in-hospital clinical deterioration (defined as any requirement of ventilatory support or critical care, or death) among consecutively hospitalised adults with highly suspected or confirmed COVID-19 who were prospectively recruited to the International Severe Acute Respiratory and Emerging Infections Consortium Coronavirus Clinical Characterisation Consortium (ISARIC4C) study across 260 hospitals in England, Scotland, and Wales. Candidate predictors that were specified a priori were considered for inclusion in the model on the basis of previous prognostic scores and emerging literature describing routinely measured biomarkers associated with COVID-19 prognosis. We used internal–external cross-validation to evaluate discrimination, calibration, and clinical utility across eight National Health Service (NHS) regions in the development cohort. We further validated the final model in held-out data from an additional NHS region (London).



Lancet Respir Med 2021

Published Online

January 11, 2021

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See Online/Comment

[https://doi.org/10.1016/S2213-2600\(21\)00006-0](https://doi.org/10.1016/S2213-2600(21)00006-0)

*Contributed equally

Institute for Global Health

(R K Gupta MRCP,

Prof I Abubakar FRCP), MRC

Clinical Trials Unit, Institute of

Clinical Trials and Methodology

UCL, London, UK

Example

- COVID-19 prognosis hospitalized patients

- **Composite outcome:** “deterioration” (in-hospital death, ventilator support, ICU)

A priori expectations

- **Event fraction** at least 30%
- 40 candidate **predictor parameters**
- **C-statistic** of 0.71 (conservative est)
-> Cox-Snell R^2 of 0.24

Development and validation of the ISARIC 4C Deterioration model for adults hospitalised with COVID-19: a prospective cohort study

Rishi K Gupta, Ewen M Harrison, Antonia Ho, Annemarie B Docherty, Stephen R Knight, Maarten van Smeden, Ibrahim Abubakar, Marc Lipman, Matteo Quartagno, Riinu Pius, Iain Buchan, Gail Carson, Thomas M Drake, Jake Dunning, Cameron J Fairfield, Carrol Gamble, Christopher A Green, David A Hare, Iain Hargrett-White, Iain Jackson, Kenneth A Mclean, Laura Merson, Jonathan S Nguyen-Van-Tam, Lisa Norman, Paul L Olliver, Martin D Poole, Clark D Russell, James Scott-Brown, Catherine A Shaw, Aziz Sheikh, Tom Solomon, Cathie Sudlow, Olivia V Swann, Liane Turtie, Peter J M Openshaw, J Kenneth Baile*, Malcolm G Semple*, Mahdad Noursadeghi*, on behalf of the ISARIC4C Investigators

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*Contributed equally

Institute for Global Health

(R K Gupta MRCP,

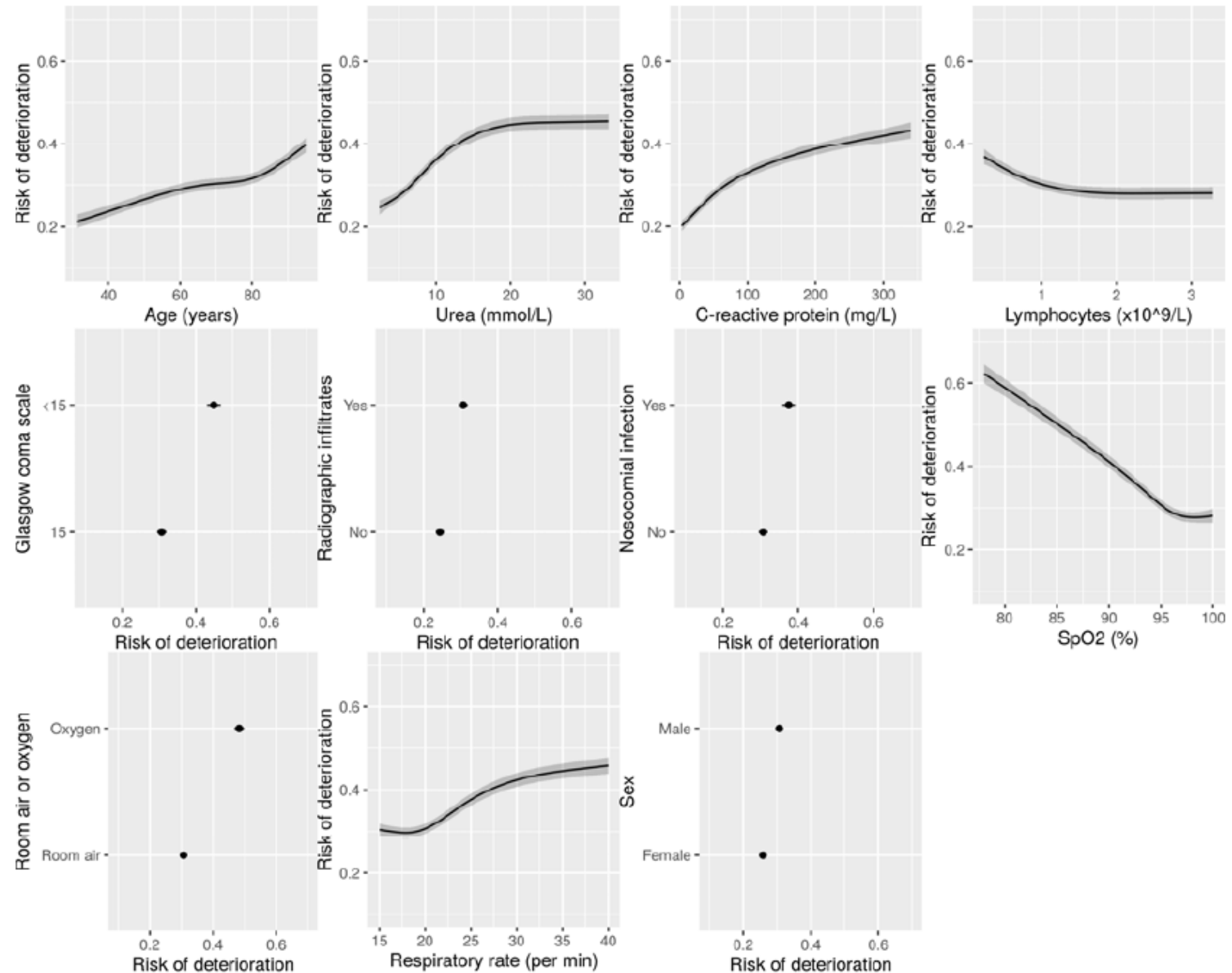
Prof I Abubakar FRCP), MRC

Clinical Trials Unit, Institute of

Clinical Trials and Methodology

Restricted cubic splines
with 4 knots: 3 degrees of
freedom

Note: EPV rule also
calculates **degrees of**
freedom of candidate
predictors, not variables!



Calculate required sample size

Criterion 1. Shrinkage: expected heuristic shrinkage factor, $S \geq 0.9$
(calibration slope, target < 10% **overfitting**)

Criterion 2. Optimism: Cox-Snell R^2 apparent - Cox-Snell R^2 validation < 0.05
(**overfitting**)

Criterion 3: A small margin of error in overall risk estimate < 0.05 absolute error
(precision estimated baseline **risk**)

(Criterion 4: a small margin of absolute error in the estimated **risks**)

Calculation

R code:

```
> require(pmsampsize)  
> pmsampsize(type="b",rsquared=0.24,parameters=40,prevalence=0.3)
```

```
NB: Assuming 0.05 acceptable difference in apparent & adjusted R-squared  
NB: Assuming 0.05 margin of error in estimation of intercept  
NB: Events per Predictor Parameter (EPP) assumes prevalence = 0.3
```

	Samp_size	Shrinkage	Parameter	Rsq	Max_Rsq	EPP
Criteria 1	1290	0.900	40	0.24	0.71	9.68
Criteria 2	962	0.871	40	0.24	0.71	7.21
Criteria 3	323	0.900	40	0.24	0.71	2.42
Final	1290	0.900	40	0.24	0.71	9.68

```
Minimum sample size required for new model development based on user inputs = 1290,  
with 387 events (assuming an outcome prevalence = 0.3) and an EPP = 9.68
```

A few alternative scenarios

- $rsquared=0.24, parameters=40, prevalence=0.3 \rightarrow EPV \geq 9.7$
- $rsquared=0.12, parameters=40, prevalence=0.3 \rightarrow EPV \geq 21.0$
- $rsquared=0.12, parameters=40, prevalence=0.5 \rightarrow EPV \geq 35.0$
- $rsquared=0.36, parameters=40, prevalence=0.2 \rightarrow EPV \geq 5$

The sample size that meets all criteria is the **MINIMUM** required

- **Why minimum?** Other criteria may be important. e.g. missing data, clustering, variable selection

Future directions

- More evidence/guidance to choose values for the sample size criteria
- Sample size for validation (for continuous outcome just published)
- Sample size for different outcomes (e.g. multi-category)
- Sample size taking into account variable selection
- Simulation based approaches
- High-dimensional data?

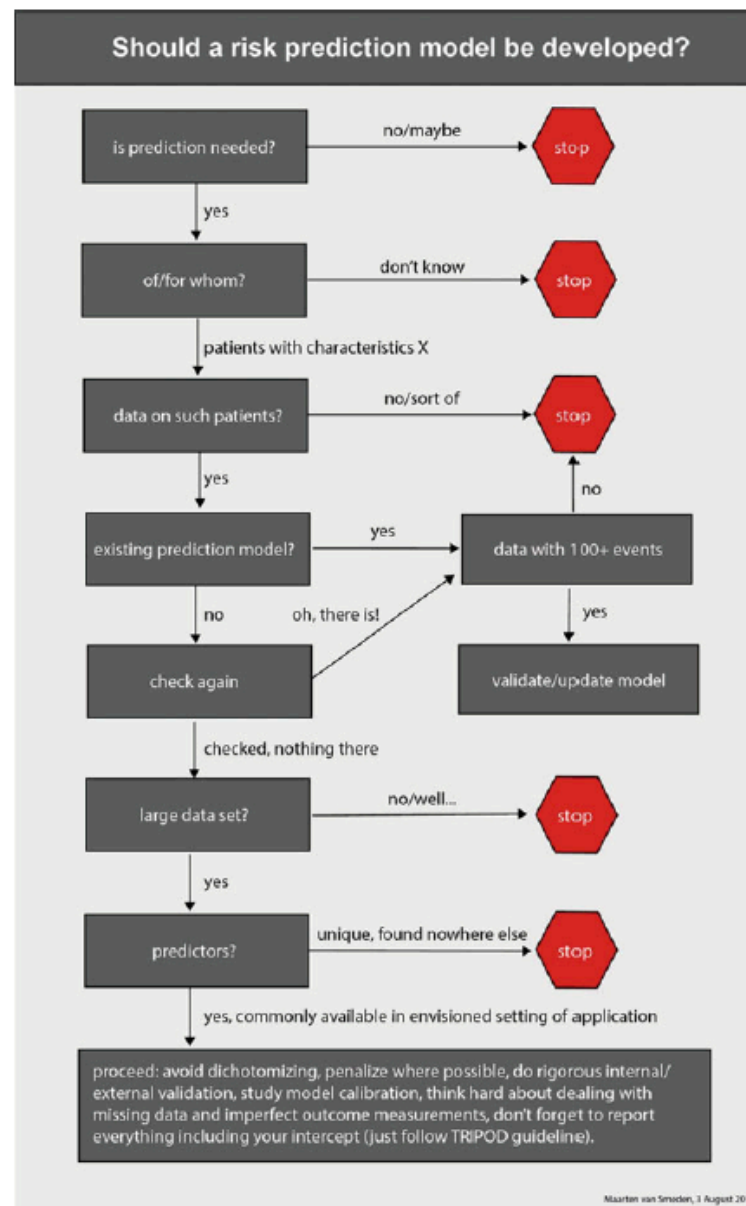


Fig. 24.6 Maarten van Smeden's flowchart for the question: should a new prediction model be developed? [633]

Work in collaboration with:

- Carl Moons
- Hans Reitsma
- Ben Van Calster (Leuven)
- Laure Wynants (Maastricht)
- Richard Riley (Keele, materials for this presentation)
- Gary Collins (Oxford, materials for this presentation)
- Ewout Steyerberg (Leiden)
- Many others



Contact: M.vanSmeden@umcutrecht.nl

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